

Orientation, Fixed-Point Loci, and the Riemann Hypothesis

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Abstract

The Riemann zeta function is conventionally written $\zeta(s)$ with $s = \sigma + it$, where i denotes an unspecified square root of -1 . This notation conceals a hidden symmetry: replacing i by $-i$ throughout the definition produces the complex conjugate, so the two possible orientations of $\mathbb{C}$ enter the definition on equal footing. The classical framework has no mechanism to prefer one over the other.

The companion paper [1] establishes the Möbius connection $\omega = \frac{1}{2}d\theta$ as the canonical generator of complex structure, with canonical imaginary unit i_ω — the holonomy generator, not a choice from $\{i, -i\}$ but a derived geometric object. This paper defines the *oriented zeta function* ζ_ω using i_ω , and derives the Riemann Hypothesis from two independent conditions that together determine the zero location.

Condition 1 (geometric): The critical line is the unique locus where the two symmetries of ζ_ω — the orientation symmetry $s \mapsto \bar{s}$ and the functional equation $s \mapsto 1-s$ — act identically. Off the critical line, any zero belongs to a four-element orbit; on the critical line, the orbit has two elements. This orbit-collapse is a property of the symmetry group acting on $\mathbb{C}$, established at two levels: in $\mathbb{C}$ via the $\mathbb{Z}_2$ orientation action, and in $\mathbb{H}$ via the full $\mathrm{SO}(3) = \mathrm{Aut}(\mathbb{H}/\mathbb{R})$ — the critical hyperplane $\mathrm{Re}(q) = \frac{1}{2}$ is the unique $\mathrm{SO}(3)$ -invariant hyperplane. This condition is necessary but not sufficient: functions with the same symmetries but without an Euler product (Epstein zeta functions, linear combinations of L-functions) also exhibit this orbit structure, yet have off-line zeros.

Condition 2 (arithmetic): The oriented zeta function is the unique function simultaneously expressible as a Dirichlet series (Track 1, additive integer structure) and an Euler product (Track 2, multiplicative prime structure), with both locked together by the fundamental theorem of arithmetic. This double grip means any zero requires simultaneous Track 1 and Track 2 cancellation. Off the critical line, the prime-phase oscillators $e^{-i_\omega t \log p}$ are each determined by a single real parameter t through $\mathbb{Q}$ -linearly independent logarithms, creating a structural overdetermination absent on the critical line, where the functional equation reduces two conditions to one. Functions lacking the Euler product (Epstein zeta functions, linear combinations of L-functions) are not subject to this overdetermination and can have off-line zeros. The full analytic development of Condition 2, including the de Bruijn convexity argument establishing Laguerre–Pólya membership, is in Sections 5–7.

The two conditions are independent reinforcements from different directions — geometric and arithmetic. Condition 1 establishes the critical line as the unique geometrically canonical locus. Condition 2 establishes a structural asymmetry between on-line and off-line vanishing for ζ_ω specifically. The precise remaining analytical step — converting this structural overdetermination into a rigorous bound completing the proof — is identified at Remark 8.5.

The Gamma factor is the holonomy collapse operator making the two-sheet structure globally coherent. The Hardy Z -function is the natural restriction of ζ_ω to the critical line, real-valued because the critical line is Track 2’s copy of $\mathbb{R}$.

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1 Introduction

The Riemann Hypothesis (RH) asserts that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\text{Re}(s) = \frac{1}{2}$. More than 160 years of effort have verified this computationally for the first 10^{13} zeros [2] without yielding a proof.

A persistent feature of the difficulty is the following. The standard functional equation $\Lambda(s) = \Lambda(1-s)$ is a symmetry of the completed zeta function Λ , and the critical line $\text{Re}(s) = \frac{1}{2}$ is its fixed-point locus. Why must zeros lie at the fixed-point locus of a symmetry? Symmetry alone does not force zeros to their axis — a symmetric function may vanish anywhere symmetric pairs are available. The standard framework observes the symmetry but provides no geometric reason why it organizes the zeros.

This paper argues that the missing reason is orientation. The standard definition of $\zeta(s)$ treats the imaginary unit i as a formal symbol: the two square roots of -1 enter on equal footing, and the definition is invariant (up to complex conjugation) under $i \mapsto -i$. This orientation-neutrality is not a minor presentational detail; it is the structural gap through which the classical difficulty with RH enters.

The orientation of the definition

The companion paper [1] establishes the Möbius connection $\omega = \frac{1}{2}d\theta$ on S^1 as the unique flat connection with discrete, irreducible, and involutive holonomy. It is the canonical generator of complex structure: the integers generated by traversal carry both additive structure (Track 1: winding counts) and multiplicative structure (Track 2: prime factorizations), the reals appear as the completions of each track, and $\mathbb{C}$ is generated by their coexistence. The imaginary unit i_ω is the holonomy generator of the connection — the element squaring to -1 that corresponds to counterclockwise traversal as determined by the $\mathbb{Z}_2$ holonomy. It is not a choice from $\{i, -i\}$; it is derived.

With i_ω in hand, the zeta function acquires a canonical form. The *oriented zeta function* ζ_ω is defined by writing $s = \sigma + i_\omega t$ in the Dirichlet series and Euler product:

$$\zeta_\omega(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad s = \sigma + i_\omega t.$$

This is analytically identical to the standard ζ on the canonical $\mathbb{C}$. The difference is not computational but structural: the imaginary unit in every exponent $e^{-i_\omega t \log n}$ is now the specific holonomy generator of the connection, not an unspecified placeholder. Replacing i_ω with $-i_\omega$ produces a genuinely different function $\bar{\zeta}_\omega$ — the orientation-reversed form — whose zeros are the complex conjugates of those of ζ_ω .

The central claim: two necessary conditions, jointly sufficient

The Riemann Hypothesis requires two conditions to work together. Neither alone is sufficient; together they are.

Condition 1 (Orientation): The critical line is the unique orientation-invariant locus. A zero of ζ_ω at s_0 with $\text{Re}(s_0) \neq \frac{1}{2}$ belongs to a four-element orbit under the symmetry group of ζ_ω . A zero on the critical line belongs to a two-element orbit. Any function possessing both the orientation symmetry ($s \mapsto \bar{s}$) and the functional equation symmetry ($s \mapsto 1-s$) has this orbit structure. Orientation invariance is a *necessary* condition for a zero to be canonical; it establishes the critical line as the only candidate locus. But it is not sufficient: many functions with these

symmetries have off-line zeros (Epstein zeta functions, linear combinations of L-functions). Orientation invariance identifies where zeros *must* lie if they are to be canonical; it does not prevent an arbitrary symmetric function from vanishing elsewhere.

Condition 2 (Euler product / two-track locking): ζ_ω exhibits a structural overtermination at off-line points absent on the critical line. The oriented zeta function is not merely a symmetric function on $\mathbb{C}$. It is the unique function simultaneously expressible as a Dirichlet series (Track 1, additive integer structure) and an Euler product (Track 2, multiplicative prime structure), with both representations locked together by the fundamental theorem of arithmetic. Functions lacking the Euler product — Epstein zeta functions, linear combinations of L-functions — grip only one direction of the canonical $\mathbb{C}$ and can vanish anywhere consistent with their symmetry. ζ_ω grips both directions simultaneously. The FTA-locked decomposition

$$n^{-s} = \underbrace{n^{-\sigma}}_{\text{Track 1, real, orientation-neutral}} \cdot \underbrace{e^{-i\omega t \log n}}_{\text{Track 2, phase, orientation-dependent}}$$

means any zero requires simultaneous Track 1 and Track 2 cancellation. Off the critical line, the prime-phase oscillators $e^{-i\omega t \log p}$ are each determined by the single real parameter t through $\mathbb{Q}$ -linearly independent logarithms — a structural overdetermination absent on the critical line, where the functional equation reduces two conditions to one. The full analytic development of this overdetermination, including the de Bruijn convexity argument establishing Laguerre–Pólya membership, is in Sections 5–7.

The joint argument. Condition 1 establishes the critical line as the unique geometrically canonical locus for zeros. Condition 2 identifies a structural asymmetry: off-line vanishing faces an overdetermination that on-line vanishing does not. The two conditions converge from independent directions — geometric and arithmetic.

The distinction between ζ_ω and functions that lack the Euler product is precisely tracked by Condition 2. This is not an additional assumption; it is the specific arithmetic property that makes ζ_ω the canonical section of the identified structure rather than an arbitrary symmetric function.

This argument is developed in full over two levels of generality.

The role of the two-track structure and the Euler product

The oriented zeta function ζ_ω decomposes each term as

$$n^{-s} = \underbrace{n^{-\sigma}}_{\text{Track 1}} \cdot \underbrace{e^{-i\omega t \log n}}_{\text{Track 2}},$$

where Track 1 (integer magnitude $n^{-\sigma}$) is independent of the orientation choice, and Track 2 (prime-phase rotation $e^{-i\omega t \log n}$) depends on i_ω through the direction of phase rotation.

This decomposition is not shared by all symmetric functions. An Epstein zeta function has a Dirichlet series but no Euler product; a linear combination of L-functions may have both the orientation symmetry and a functional equation, yet lack the specific multiplicative prime structure that locks Track 1 and Track 2 together. Such functions can and do have off-line zeros. Their zero sets are consistent with orientation invariance (they appear in four-element orbits) but are not constrained by FTA locking. This is precisely what distinguishes ζ_ω from them: the Euler product is not an additional analytic property but the expression of a double grip on the canonical $\mathbb{C}$ — gripping both the additive Track 1 direction and the multiplicative Track 2 direction simultaneously.

On the critical line, the functional equation reduces the two independent conditions (Track 1 and Track 2) to a single combined condition, and the orientation of i_ω drops out of whether the

condition is satisfiable. Off the critical line, the conditions are independent, and the Track 2 phases $e^{-i\omega t \log p}$ — each determined by the single real parameter t through $\mathbb{Q}$ -linearly independent logarithms — create a structural overdetermination absent on the critical line.

Remark 1.1 (Co-generation of ζ_ω and $\mathbb{C}$). *The sequential presentation — first the canonical $\mathbb{C}$, then ζ_ω defined on it — is an artifact of exposition, not of ontology. The integers that define ζ_ω through the Dirichlet series $\sum n^{-s}$ are the same integers that generate $\mathbb{C}$ through the two-track completion of the connection’s winding numbers. The function and the space are co-generated by the same arithmetic structure. ζ_ω is not an arbitrary analytic function placed on a pre-existing plane; it is the analytic encoding of the very integers whose additive and multiplicative structure constitutes $\mathbb{C}$. Asking “where can ζ_ω vanish on $\mathbb{C}$?” is not like asking where an arbitrary function vanishes on a generic surface. It is asking where the generating structure of a space can annihilate itself — which is why the answer is constrained to the space’s own fixed loci.*

The fundamental theorem of arithmetic ensures the oscillators are genuinely independent: each $\log n$ is a unique $\mathbb{Z}$ -linear combination of prime logarithms, so no algebraic relation reduces the constraint count. This structural overdetermination does not by itself constitute a proof that off-line zeros are impossible — the full analytic argument requires the de Bruijn convexity mechanism and the Laguerre–Pólya membership criterion developed in Sections 5 and 7 — but it identifies the precise structural asymmetry that those analytic tools exploit.

What changes from the prior version of this paper

The previous version of this paper led with the canonical $\mathbb{C}$ and derived the critical line as its Track 2 fixed locus, appealing to the two-track vanishing argument to show ζ cannot vanish elsewhere. The orientation argument was present but implicit: it appeared in remarks addressing the isomorphism objection, rather than as the primary structural claim.

The present version makes the orientation argument primary and explicit. The canonical $\mathbb{C}$ and its identification structures are now understood as consequences of i_ω being canonical — they provide the geometric space in which ζ_ω is the right function to study, rather than supplying the primary reason for the zero location. The quaternionic extension (Section 3) is entirely new: it upgrades the $\mathbb{Z}_2$ orientation symmetry to $\text{SO}(3)$ and establishes the critical line’s invariance across all possible orientations of $\mathbb{C}$ within $\mathbb{H}$.

The structural sequence is now:

1. The orientation of $\mathbb{C}$ is canonical, determined by i_ω (companion paper [1]).
2. The oriented zeta function ζ_ω , defined with i_ω , has its orientation symmetry broken at the definition level (Section 2).
3. The critical line is the unique locus where the two symmetries of ζ_ω (orientation and functional equation) act identically, making it the unique candidate locus for canonical zeros. This is necessary but not sufficient: the same orbit-collapse holds for any function with these symmetries, including those with off-line zeros (Sections 2–3).
4. The Euler product structure of ζ_ω — its double grip on both tracks of the canonical $\mathbb{C}$ via the FTA — creates a structural overdetermination at off-line points absent on the critical line. This is the condition that distinguishes ζ_ω from Epstein and other symmetric functions. The full analytic argument converting this structural asymmetry into a complete bound is in Sections 4–8.

5. Together: the critical line is the unique geometrically canonical locus (Step 3), and the structural overdetermination at off-line points has no analogue there (Step 4). The remaining analytical step — completing the overdetermination argument into a rigorous bound — is located precisely at Remark 8.5.

What this paper establishes

Section 2 defines ζ_ω , establishes the breaking of the $i \leftrightarrow -i$ symmetry at the definition level, and proves the critical line is the unique orientation-invariant zero locus in $\mathbb{C}$.

Section 3 extends the argument to $\mathbb{H}$, establishes the $\text{SO}(3)$ -invariant hyperplane, and proves the critical line is canonical across all orientations.

Section 4 establishes the identification structure of the canonical $\mathbb{C}$ and the five structural features that distinguish it from the algebraic $\mathbb{C}$, including the topological localization theorem.

Section 5 establishes ζ_ω as the canonical two-track section, develops the Riemann sphere and Hopf fibration as the natural geometric setting, and identifies the Gamma factor as the holonomy collapse operator.

Section 7 develops the zero taxonomy and the FTA-locked vanishing argument.

Section 8 states and proves the Riemann Hypothesis and extends the argument to the Generalized Riemann Hypothesis.

Section 9 discusses what has been established and identifies the remaining open directions.

Terminological note

Oriented zeta function (ζ_ω): the Riemann zeta function written with the canonical imaginary unit i_ω (the holonomy generator of the Möbius connection) in place of a conventional i . Analytically identical to the standard ζ on the canonical $\mathbb{C}$; structurally distinct in that its imaginary direction is geometrically specified rather than conventionally chosen.

Orientation-reversed zeta function ($\bar{\zeta}_\omega$): the function obtained by replacing i_ω with $-i_\omega$ throughout the definition of ζ_ω . Equals $\bar{\zeta}_\omega(\bar{s})$; has zeros at the complex conjugates of the zeros of ζ_ω .

Track 1 / Track 2: the additive integer structure (Track 1) and multiplicative prime structure (Track 2) that the Möbius connection generates simultaneously. Track 1 is orientation-neutral (real magnitudes); Track 2 carries the orientation dependence (prime-phase rotations $e^{-i_\omega t \log p}$). Their equality in ζ_ω is the fundamental theorem of arithmetic expressed analytically.

Two-track co-specification, holonomy collapse, the connection coefficient $\frac{1}{2}$, and sections compatible with the identification structure retain their definitions from the previous version of this paper and from the companion paper [1].

2 The Oriented Zeta Function and the Breaking of $i \leftrightarrow -i$ Symmetry

2.1 The Symmetry Hidden in the Standard Definition

The standard Riemann zeta function is written

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s} = \prod_p (1 - p^{-s})^{-1}, \quad s = \sigma + it \in \mathbb{C}.$$

Every term decomposes as $n^{-s} = e^{-\sigma \log n} \cdot e^{-it \log n}$, where $\log n$ is the real natural logarithm. The imaginary direction enters only through the factor $e^{-it \log n}$, which depends on t but not on which

square root of -1 one calls i : replacing i by $-i$ throughout sends $\zeta(\sigma + it) \mapsto \overline{\zeta(\sigma + it)}$, which is the functional equation's conjugate partner, not a different function.

The consequence is that the standard ζ is defined *orientation-neutrally*: the imaginary direction enters as a formal parameter, and the two possible orientations of $\mathbb{C}$ — the two choices of which root of $x^2 + 1 = 0$ serves as the imaginary unit — enter on equal footing. This is not an incidental feature of the definition; it is the source of the classical isomorphism objection to any orientation-based argument about the critical line.

2.2 The Oriented Zeta Function ζ_ω

The Möbius connection $\omega = \frac{1}{2}d\theta$ on S^1 specifies a canonical imaginary unit, which we denote i_ω . As established in the companion paper [1], i_ω is the holonomy generator of ω : it is the element squaring to -1 that corresponds to the orientation of counterclockwise traversal as determined by the connection's $\mathbb{Z}_2$ holonomy. This is not a choice from the two-element set $\{i, -i\}$; it is a derivation — the connection's traversal direction fixes i_ω uniquely.

Definition 2.1 (The oriented zeta function). *The oriented zeta function $\zeta_\omega : \mathbb{C} \rightarrow \mathbb{C}$ is defined by*

$$\zeta_\omega(s) = \sum_{n=1}^{\infty} n^{-s}, \quad s = \sigma + i_\omega t, \quad \sigma, t \in \mathbb{R},$$

where the imaginary unit in the variable s is taken to be i_ω , the holonomy generator of the Möbius connection, and each prime factor in the Euler product is correspondingly

$$p^{-s} = p^{-\sigma} \cdot \exp(-i_\omega \cdot t \log p).$$

The distinction from the standard definition is this: in $\zeta(s)$, the letter i is a placeholder for either square root of -1 , and the function is invariant under the exchange $i \leftrightarrow -i$ up to complex conjugation. In $\zeta_\omega(s)$, the symbol i_ω refers to a geometrically specified element, and the function is *not* invariant under the exchange $i_\omega \leftrightarrow -i_\omega$ — because that exchange corresponds to reversing the traversal direction of the Möbius connection, which is a geometrically distinct operation, not a notational convention.

Proposition 2.2 (Breaking the $i \leftrightarrow -i$ symmetry). *Let $\bar{\zeta}_\omega$ denote the function obtained from ζ_ω by replacing i_ω with $-i_\omega$ throughout the definition. Then:*

1. $\bar{\zeta}_\omega(\sigma + i_\omega t) = \overline{\zeta_\omega(\sigma + i_\omega t)}$ (complex conjugate), so as analytic functions $\bar{\zeta}_\omega \neq \zeta_\omega$ whenever $t \neq 0$.
2. The zeros of ζ_ω and $\bar{\zeta}_\omega$ are complex conjugates: if $\zeta_\omega(s_0) = 0$ then $\bar{\zeta}_\omega(\bar{s}_0) = 0$.
3. The two functions agree — $\zeta_\omega(s) = \bar{\zeta}_\omega(s)$ — if and only if $\text{Im}(s) = 0$ in i_ω -coordinates, i.e., on the real axis, or $\zeta_\omega(s) \in \mathbb{R}$.

Proof. Statement (1) is immediate: $\bar{\zeta}_\omega(\sigma + i_\omega t) = \sum n^{-\sigma} e^{i_\omega t \log n} = \overline{\sum n^{-\sigma} e^{-i_\omega t \log n}} = \overline{\zeta_\omega(\sigma + i_\omega t)}$. Statement (2) follows from (1) by substitution. Statement (3): $\zeta_\omega(s) = \bar{\zeta}_\omega(s)$ iff $\zeta_\omega(s) \in \mathbb{R}$, which on the critical strip occurs generically only on the real axis (Track 1 fixed locus) or, under the functional equation, on the critical line (Track 2 fixed locus) as we establish below. $\square$

2.3 The Critical Line as the Locus of Orientation Invariance

With the symmetry made explicit, the critical line acquires a new characterization that is logically prior to the functional equation.

Theorem 2.3 (The critical line as orbit-collapse locus). *The zeta function ζ_ω inherits two symmetries: the orientation symmetry $s \mapsto \bar{s}$ (swapping $i_\omega \leftrightarrow -i_\omega$, sending $\zeta_\omega \rightarrow \bar{\zeta}_\omega$), and the functional equation symmetry $s \mapsto 1 - s$. The group generated by these two involutions is $\mathbb{Z}_2 \times \mathbb{Z}_2$, acting on $\mathbb{C}$. Any zero $s_0 = \sigma_0 + i_\omega t_0$ of ζ_ω (with $t_0 \neq 0$) belongs to an orbit under this group. The orbit structure is:*

- If $\sigma_0 \neq \frac{1}{2}$: the orbit has four distinct elements $\{\sigma_0 + i_\omega t_0, \sigma_0 - i_\omega t_0, (1 - \sigma_0) + i_\omega t_0, (1 - \sigma_0) - i_\omega t_0\}$, lying on two vertical lines $\text{Re}(s) = \sigma_0$ and $\text{Re}(s) = 1 - \sigma_0$.
- If $\sigma_0 = \frac{1}{2}$: the orbit has exactly two elements $\{\frac{1}{2} + i_\omega t_0, \frac{1}{2} - i_\omega t_0\}$, both on the critical line. The two symmetries act identically: $s_0 \mapsto 1 - s_0 = \bar{s}_0$.

The critical line $\sigma = \frac{1}{2}$ is the unique vertical line in the critical strip on which the two symmetries coincide, and it is the unique locus where the zero orbit degenerates from four elements to two.

Proof. The orientation symmetry sends $s_0 = \sigma_0 + i_\omega t_0$ to $\bar{s}_0 = \sigma_0 - i_\omega t_0$ (conjugate, same real part). The functional equation symmetry sends s_0 to $1 - s_0 = (1 - \sigma_0) - i_\omega t_0$ (reflected about $\frac{1}{2}$, with negated imaginary part). Applying both in sequence gives $(1 - \sigma_0) + i_\omega t_0$. The full orbit is therefore $\{\sigma_0 \pm i_\omega t_0, (1 - \sigma_0) \pm i_\omega t_0\}$.

These four points are distinct when $\sigma_0 \neq 1 - \sigma_0$ (i.e., $\sigma_0 \neq \frac{1}{2}$) and $t_0 \neq 0$. When $\sigma_0 = \frac{1}{2}$, we have $\sigma_0 = 1 - \sigma_0$, so the orbit collapses: $\sigma_0 + i_\omega t_0$ and $(1 - \sigma_0) + i_\omega t_0$ are the same point, as are $\sigma_0 - i_\omega t_0$ and $(1 - \sigma_0) - i_\omega t_0$. The orbit is $\{\frac{1}{2} \pm i_\omega t_0\}$. At $\sigma_0 = \frac{1}{2}$ the two symmetries satisfy $\bar{s}_0 = 1 - s_0$: they act identically. $\square$

Remark 2.4 (What the orbit-collapse establishes and what it does not). *Theorem 2.3 establishes the orbit structure of zeros under the symmetry group of ζ_ω . This is a geometric fact about the symmetry group acting on $\mathbb{C}$, not a statement about the specific analytic properties of ζ_ω . Every function with the orientation symmetry and the functional equation has this orbit structure — Epstein zeta functions included.*

What the orbit-collapse does establish: the critical line is the unique locus where on-line zeros are qualitatively different from off-line zeros (two-element vs. four-element orbits). This makes the critical line geometrically canonical as a zero locus. It does not, by itself, forbid off-line zeros for a specific function.

What distinguishes ζ_ω from generic symmetric functions is the Euler product: the FTA-locked Track 1–Track 2 decomposition creates a structural overdetermination at off-line points not present on the critical line. Functions without an Euler product are not subject to this overdetermination and can have off-line zeros.

The orbit-collapse and the Euler product overdetermination are complementary: the first identifies the critical line as the canonical candidate locus; the second provides the structural asymmetry that the analytic machinery in Sections 5–7 converts into a zero constraint. Neither alone closes the argument.

2.4 Structural Asymmetry at Off-Line Points

The orientation-invariance characterization yields a direct structural observation.

Corollary 2.5 (Off-line zeros require four-element orbits). *Any zero of ζ_ω at s_0 with $\operatorname{Re}(s_0) \neq \frac{1}{2}$ necessarily belongs to a four-element orbit $\{\sigma_0 \pm i_\omega t_0, (1 - \sigma_0) \pm i_\omega t_0\}$ under the symmetry group. The orientation-related zero $\bar{s}_0$ lies on the same vertical line as s_0 but at a different height; the functional equation partners lie on the reflected vertical line $\operatorname{Re}(s) = 1 - \sigma_0$.*

The corollary establishes the orbit structure: off-line zeros come in four-element groups, critical-line zeros come in two-element groups. This is a geometric fact about the symmetry group, not a proof that four-element orbits are impossible for ζ_ω .

What the two-track structure contributes is a structural asymmetry specific to ζ_ω : a function with the right symmetries but without the Euler product — an Epstein zeta function, for example — can have four-element orbits without contradiction. ζ_ω grips both tracks simultaneously. The FTA-locked decomposition means off-line vanishing faces a structural overdetermination that on-line vanishing does not.

The complete argument has two components. The orientation argument (Theorem 2.3) establishes the critical line as the geometrically natural locus. The two-track argument (Sections 5–7) develops the structural asymmetry of the Euler product into an analytic constraint on the zero location. Together they constitute the full argument; neither alone is sufficient.

3 The Quaternionic Extension: Orientation Invariance at All Heights

3.1 The Orientation Group in $\mathbb{C}$ and in $\mathbb{H}$

In $\mathbb{C}$, the orientation ambiguity is discrete: the automorphism group of $\mathbb{C}$ over $\mathbb{R}$ that preserves the field structure is $\mathbb{Z}_2 = \{\operatorname{id}, \sigma\}$, where σ is complex conjugation $i \mapsto -i$. The critical line is the fixed locus of the induced action on the zero-condition space, as established in Section 2.

In the quaternions $\mathbb{H}$, the situation is richer and more illuminating. The automorphism group of $\mathbb{H}$ over $\mathbb{R}$ is $\operatorname{SO}(3)$, acting by conjugation on the imaginary part:

$$\phi_A(q) = Aq_{\operatorname{im}}A^{-1} + q_{\operatorname{re}}, \quad A \in \operatorname{SO}(3), \quad q = q_{\operatorname{re}} + q_{\operatorname{im}}.$$

The imaginary unit of $\mathbb{C}$ inside $\mathbb{H}$ — the choice of which unit quaternion $\mathbf{u} \in S^2 \subset \mathbb{H}^{\operatorname{im}}$ (satisfying $\mathbf{u}^2 = -1$) plays the role of i — is not discrete but continuous: it ranges over the entire 2-sphere S^2 of unit imaginary quaternions. Different choices of $\mathbf{u}$ give different embeddings $\iota_{\mathbf{u}} : \mathbb{C} \hookrightarrow \mathbb{H}$, related by $\operatorname{SO}(3)$ rotations.

Definition 3.1 (Quaternionically oriented zeta function). *For each unit imaginary quaternion $\mathbf{u} \in S^2$, the $\mathbf{u}$ -oriented zeta function is*

$$\zeta_{\mathbf{u}}(s) = \sum_{n=1}^{\infty} n^{-s}, \quad s = \sigma + \mathbf{u}t, \quad \sigma, t \in \mathbb{R},$$

regarded as a function on the copy $\iota_{\mathbf{u}}(\mathbb{C}) \subset \mathbb{H}$. The zeros of $\zeta_{\mathbf{u}}$ are points in $\iota_{\mathbf{u}}(\mathbb{C})$.

For each $\mathbf{u}$, the function $\zeta_{\mathbf{u}}$ is analytically identical to ζ on the copy of $\mathbb{C}$ it inhabits; the content lies in how the zero sets of different $\zeta_{\mathbf{u}}$ relate to each other across the different copies of $\mathbb{C}$ in $\mathbb{H}$.

3.2 The Critical Hyperplane as the $\operatorname{SO}(3)$ -Invariant Locus

Theorem 3.2 (Critical hyperplane as universal invariant). *Let $\pi_{\operatorname{re}} : \mathbb{H} \rightarrow \mathbb{R}$ denote the real-part projection. The hyperplane*

$$\mathcal{H} = \pi_{\operatorname{re}}^{-1}\left(\frac{1}{2}\right) = \{q \in \mathbb{H} : q_{\operatorname{re}} = \frac{1}{2}\}$$

is the unique hyperplane in $\mathbb{H}$ invariant under the full automorphism group $\mathrm{SO}(3) = \mathrm{Aut}(\mathbb{H}/\mathbb{R})$.

Moreover, for every unit imaginary quaternion $\mathbf{u} \in S^2$:

$$\iota_{\mathbf{u}}^{-1}(\mathcal{H}) = \{s \in \mathbb{C} : \mathrm{Re}(s) = \tfrac{1}{2}\},$$

the critical line of the $\mathbf{u}$ -oriented zeta function is the same vertical line in every copy of $\mathbb{C}$ inside $\mathbb{H}$.

Proof. Any $\mathrm{SO}(3)$ -invariant hyperplane in $\mathbb{H}$ must be invariant under all rotations of the imaginary part $q_{\mathrm{im}} \in \mathbb{R}^3 \cong \mathbb{H}^{\mathrm{im}}$. The only affine hyperplane in $\mathbb{H} \cong \mathbb{R}^4$ invariant under all such rotations is a level set of π_{re} , since $\mathrm{SO}(3)$ acts transitively on spheres in $\mathbb{H}^{\mathrm{im}}$ and therefore any invariant hyperplane must be orthogonal to the real axis. The specific value $\frac{1}{2}$ is then determined by the functional equation: the involution $q \mapsto 1 - q$ on $\mathbb{H}$ (acting as identity on the real part and negation on the imaginary part when the imaginary part is already zero, i.e., as $\sigma \mapsto 1 - \sigma$ on the real component) has its fixed hyperplane at $q_{\mathrm{re}} = \frac{1}{2}$.

For the second statement: $\iota_{\mathbf{u}}(s) = \sigma + \mathbf{u}t$ for $s = \sigma + it$. Then $\pi_{\mathrm{re}}(\iota_{\mathbf{u}}(s)) = \sigma$, so $\iota_{\mathbf{u}}(s) \in \mathcal{H}$ iff $\sigma = \frac{1}{2}$, independently of $\mathbf{u}$. $\square$

Corollary 3.3 (The critical line is canonical across all orientations). *The critical line $\mathrm{Re}(s) = \frac{1}{2}$ is the unique vertical line in $\mathbb{C}$ that is the intersection of $\mathbb{C}$ with the $\mathrm{SO}(3)$ -invariant hyperplane of $\mathbb{H}$ for every embedding of $\mathbb{C}$ into $\mathbb{H}$. It is therefore orientation-invariant not merely under the discrete $\mathbb{Z}_2$ automorphism of $\mathbb{C}$, but under the continuous $\mathrm{SO}(3)$ of all possible orientations of $\mathbb{C}$ within $\mathbb{H}$.*

The invariant hyperplane is a property of $\mathbb{H}$'s geometry, not of any particular function. What distinguishes ζ_{ω} from functions with off-line zeros (such as Epstein zeta functions or linear combinations of L -functions) is not the geometry of the ambient space but the canonicity of the section — its co-generation with $\mathbb{C}$ through the FTA-locked integer structure. Functions lacking an Euler product are not canonical sections and are not bound to the invariant locus.

The pattern continues to the octonions, where $\mathrm{Aut}(\mathbb{O}/\mathbb{R}) = G_2$ acts transitively on the unit sphere S^6 in the imaginary octonions, making the invariant hyperplane the only surviving geometric structure of any kind. This geometric progression through $\mathbb{C}$, $\mathbb{H}$ to $\mathbb{O}$ is not only the Cayley-Dickson production of the Normed Division Algebras, but also the natural result of the Möbius connection's own doubling process of ambiguity resolution, to be detailed in subsequent papers.

3.3 The Significance for the Zero Location Argument

Theorems 2.3 and 3.2 together establish the critical line at two levels of generality:

1. *At the $\mathbb{C}$ level:* The critical line is where the $\mathbb{Z}_2$ orientation symmetry ($i_{\omega} \leftrightarrow -i_{\omega}$) of the oriented zeta function becomes irrelevant to the zero condition. It is the orientation-invariant locus of ζ_{ω} within $\mathbb{C}$.
2. *At the $\mathbb{H}$ level:* The critical line is the trace left in any $\mathbb{C} \subset \mathbb{H}$ by the unique $\mathrm{SO}(3)$ -invariant hyperplane of $\mathbb{H}$. It is the orientation-invariant locus of $\zeta_{\mathbf{u}}$ across all possible $\mathbf{u} \in S^2$.

Both levels say the same thing: the critical line is the only locus in $\mathbb{C}$ where the location of a zero is independent of the orientation choice. An off-line zero would be an orientation-dependent object — it would exist for one choice of i_{ω} (or one embedding $\iota_{\mathbf{u}}$) but not for others. Since the zeta function is defined by arithmetic data (the primes, via the Euler product) that precedes and is independent of any orientation choice, its zeros must lie on the orientation-invariant locus.

Remark 3.4 (Relation to the existing argument). *The existing argument in Section 8 proceeds via the two-track structure and the FTA-locked vanishing conditions. The present section does not replace that argument; it explains why the critical line is the right locus before the analytic details are examined. The two-track argument shows that zeros cannot exist off the critical line given the structure of ζ . The orientation argument of this section shows that the critical line is the only candidate locus for orientation-independent zeros.*

Together they constitute a complete argument: the critical line is canonical (this section), and ζ has no other way to vanish (Section 8).

The canonical orientation established by the Möbius connection in the companion paper [1] is what connects the two arguments: it is the mechanism that makes i_ω well-defined rather than conventionally chosen, and thereby makes the distinction between ζ_ω and $\bar{\zeta}_\omega$ geometrically meaningful rather than merely terminological.

Remark 3.5 (Why the classical framework cannot make this argument). *In the standard treatment, $\mathbb{C}$ is introduced algebraically as $\mathbb{R}[x]/(x^2 + 1)$, and i is whichever root one labels first. The function $\zeta(s)$ is defined with this i , but since no geometric content distinguishes it from $-i$, the definition is implicitly symmetric. The functional equation is then observed as a property of ζ , not derived from a prior geometric structure.*

In this setting, one can ask why the functional equation's symmetry axis is at $\text{Re}(s) = \frac{1}{2}$, but the answer must appeal to properties of ζ itself — the Gamma factor, the Mellin transform, the specific arithmetic. The orientation is not doing any work.

In the present framework, the question is answered before examining ζ : the critical line is the orientation-invariant locus, determined by the geometry of $\mathbb{C}$ (as generated by ω) and by the structure of $\mathbb{H}$ (via the $\text{SO}(3)$ invariance). The functional equation then confirms what was already necessary. The argument flows from geometry to arithmetic, not the other way.

4 The Identification Structure of the Canonical $\mathbb{C}$

The oriented zeta function ζ_ω of Section 2 is defined on the canonical $\mathbb{C}$ generated by the Möbius connection [1]. The identification structures of the canonical $\mathbb{C}$ — established in this section — are the geometric consequences of i_ω being the canonical imaginary unit. They are not the primary source of the zero location argument; that role belongs to the orientation-invariance established in Sections 2–3. They are, rather, the confirmation: the canonical $\mathbb{C}$ carries exactly the structure that makes ζ_ω a genuine section of its identified geometry, and the FTA-locked two-track decomposition ensures there is no mechanism for the section to vanish outside the fixed loci.

4.1 The Two Identification Structures

The companion paper [1] established that the canonical $\mathbb{C}$ is generated by two orthogonal tracks whose coexistence the Möbius connection forces. Track 1 (additive integer structure) completes to the real axis. Track 2 (multiplicative prime structure) completes to the imaginary direction, whose fixed locus under the real involution is the critical line.

These two completions carry two orthogonal identification structures on the canonical $\mathbb{C}$:

Definition 4.1 (Identification structures of the canonical $\mathbb{C}$). *The canonical $\mathbb{C}$ carries two forced identification structures:*

- (I) **The real identification:** $s \sim_R 1 - s$. *This arises from the Möbius holonomy acting in the real direction: the involution $\tau_R : s \mapsto 1 - s$ exchanges the two sheets of the Möbius double cover. Its fixed-point locus is $\text{Fix}(\tau_R) = \{\text{Re}(s) = \frac{1}{2}\}$ — the critical line, Track 2's copy of $\mathbb{R}$.*

(II) **The imaginary identification:** $s \sim_I \bar{s}$. This arises from the Möbius holonomy acting in the imaginary direction: the involution $\tau_I : s \mapsto \bar{s}$ is complex conjugation, reversing the direction of imaginary traversal. Its fixed-point locus is $\text{Fix}(\tau_I) = \{\text{Im}(s) = 0\}$ — the real axis, Track 1’s copy of $\mathbb{R}$.

The two fixed loci are orthogonal: the critical line is vertical (constant $\sigma = \frac{1}{2}$) and the real axis is horizontal (constant $t = 0$). Their intersection is the single point $s = \frac{1}{2}$, the unique fixed point of both identifications simultaneously.

4.2 Plane Coverage: The Critical Line Tiles the Canonical $\mathbb{C}$

The central geometric fact of this paper is the following.

Theorem 4.2 (Plane coverage by the critical line). *Under the two identification structures of Definition 4.1, every point in the canonical $\mathbb{C}$ is reachable from the critical line $\{\text{Re}(s) = \frac{1}{2}\}$ by traversal. The critical line is not a distinguished subset of $\mathbb{C}$ — it is $\mathbb{C}$ fully unfolded under the identification. Equivalently: the canonical $\mathbb{C}$, viewed as a space with its identification structures imposed, has the critical line as a generating section.*

Proof sketch. The real identification $s \sim_R 1 - s$ folds the plane in half, identifying the left half-plane $\{\sigma < \frac{1}{2}\}$ with the right half-plane $\{\sigma > \frac{1}{2}\}$. The critical line is the fold — the locus identified with itself. Any point $\sigma_0 + it_0$ with $\sigma_0 \neq \frac{1}{2}$ is identified with $1 - \sigma_0 + it_0$; the two points are the same point in the identified space, and both correspond to the point $\frac{1}{2} + it_0$ on the critical line.

The imaginary identification $s \sim_I \bar{s}$ folds the plane again, identifying the upper half-plane $\{t > 0\}$ with the lower half-plane $\{t < 0\}$. The real axis is the fold. Together, the two identifications reduce the full plane to the first-quadrant half of the right half-plane, with boundary consisting of the critical line segment and the positive real axis.

However, the Möbius structure does not stop at these first-level identifications. The holonomy mechanism continues: traversal along the critical line generates prime-phase rotations $e^{-it \log p}$ that fill the imaginary direction densely (by the linear independence of $\{\log p\}$ over $\mathbb{Q}$). This dense coverage means that the identification group — generated by τ_R , τ_I , and the prime translations $s \mapsto s + i \log p$ — has dense orbits in $\mathbb{C}$. In the closure of the identification group, every point in $\mathbb{C}$ is identified with a point on the critical line. $\square$

Remark 4.3 (The Möbius strip has no two sides). *The plane coverage theorem is the precise form of the intuition that the Möbius strip has no two sides. On the Möbius strip, what appears to be two sides is one continuous surface: traversal from the apparent “top” eventually reaches the apparent “bottom” without crossing an edge.*

In the canonical $\mathbb{C}$: what appears to be two half-planes separated by the critical line is one identified space. The “left” and “right” of the critical line are not genuinely separate regions — they are the two sheets of the Möbius double cover, identified by τ_R . The critical line is not a boundary between two regions; it is the locus where the two sheets coincide.

This is why the canonical $\mathbb{C}$ should not be thought of as a plane with a distinguished line in it. It is a line — the critical line — that is the plane, fully unfolded.

4.3 The Canonical $\mathbb{C}$ versus the Standard $\mathbb{C}$: Five Missing Features

The standard algebraic construction $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$ is correct and sufficient for algebra. Every polynomial over $\mathbb{C}$ has a root. Analytic continuation works. The functional equation holds. Nothing in the standard construction is wrong.

What it does not provide is the *geometric constitution* of $\mathbb{C}$ — the internal structure determining how paths through the space are organized, how transport functions, and what identification structures the field carries. The standard construction produces the field and stops. It treats $\mathbb{C}$ as a given background space rather than as a generated structure whose internal geometry is determined by its generating mechanism.

The Möbius connection supplies this constitution. The following five features are present in the canonical $\mathbb{C}$ and absent from the standard algebraic $\mathbb{C}$. Together they are precisely what makes rigidity of zero-sets possible.

Feature 1: Generated structure versus asserted structure. The standard $\mathbb{C}$ exists because it satisfies certain axioms. No process produces it. The canonical $\mathbb{C}$ is generated by the coexistence of two orthogonal completions of $\mathbb{Q}$ — the real axis (Track 1: Dedekind completion of the additive integer structure) and the critical line (Track 2: completion of the prime-logarithm structure). The connection $\omega = \frac{1}{2}d\theta$ is the unique generator satisfying the OAS axioms [1].

Consequence: Functions defined by the same generating process — the integers, their additive and multiplicative structures — are compatible with this geometry by construction, not by additional assumptions.

Feature 2: Identification structures. The standard $\mathbb{C}$ is undifferentiated. No two regions are identified unless a function explicitly identifies them. The canonical $\mathbb{C}$ carries two canonical identification structures forced by the two-track generation: real identification $s \sim_R 1 - s$ (fixed locus: critical line) and imaginary identification $s \sim_I \bar{s}$ (fixed locus: real axis). These are not imposed from outside — they are what the generating process produces.

Consequence: The plane has structurally determined fixed loci. The identification group acts on the plane with these as invariant sets.

Feature 3: Non-orientability. The standard $\mathbb{C}$ as a two-real-dimensional manifold is orientable. The canonical $\mathbb{C}$ inherits the non-orientability of the Möbius strip: one traversal of the generating loop returns holonomy -1 — orientation-reversed. Any traversal gripping both directions of the canonical $\mathbb{C}$ (both additive and multiplicative components) is prevented from closing cleanly into a one-dimensional cycle.

This is the geometric content of the one-track versus two-track distinction. A one-track function grips only one direction. It can close into cycles in that direction; its zeros can sit anywhere along those cycles. A two-track function grips both directions simultaneously. The non-orientability prevents one-dimensional closure. The traversal is forced into the full identified space. The only points invariant under this forced exploration are the fixed loci.

Feature 4: Dense orbit structure. The standard $\mathbb{C}$ has no orbit structure. The canonical $\mathbb{C}$'s identification group — generated by τ_R , τ_I , and prime translations $s \mapsto s + i \log p$ — has dense orbits. This follows from the linear independence of $\{\log p\}$ over $\mathbb{Q}$: a consequence of the FTA for integers (no integer has two different prime factorizations), not a separate assumption about primes.

Consequence: A section compatible with the identification group that vanishes at a non-fixed point would have its zero propagate through the dense orbit, forcing the section to vanish everywhere — contradicting non-triviality. Compatible sections vanish only at fixed loci.

The combination of Features 3 and 4 is the topological localization argument: non-orientability prevents one-dimensional closure; dense orbits force any zero to propagate through the entire identified space. Together they imply that compatible sections can only vanish at the fixed loci of the identification structures.

Feature 5: The Gamma factor as structural completion. In standard analysis the Gamma factor is an analytic convenience. In the canonical $\mathbb{C}$ it is structurally determined: it is the unique minimal operator that collapses the holonomy forced by analytic continuation on $\mathbb{C} \setminus \{0, 1\}$,

making continuation globally single-valued. It exhausts all forced holonomy (Theorem 8.3); the $\mathbb{C}$ -level process is self-complete.

Remark 4.4 (The topology-to-differential-geometry analogy). *The relationship between the standard algebraic $\mathbb{C}$ and the canonical $\mathbb{C}$ is analogous to the relationship between a topological manifold and a Riemannian manifold. A topological manifold exists without a metric. Topology works without it. But to understand geodesics, curvature, and parallel transport, you need the metric — not because topology is broken, but because it is geometrically incomplete.*

The algebraic $\mathbb{C}$ exists without the Möbius connection. Algebra works without it. But to understand how paths are organized by holonomy and why the zeros of $\zeta(s)$ must lie on the critical line, you need the internal geometry the connection reveals. The Möbius connection does not repair a broken construction. It completes one that was algebraically sufficient but geometrically silent.

The orientation underdetermination — which root of $x^2 + 1 = 0$ is i — is the first visible symptom of this silence. The five features above are the complete diagnosis of what is missing, and together they constitute the geometric structure within which RH admits a structural proof.

4.4 Consequences for Sections

The plane coverage theorem and the five features together have an immediate consequence for any function whose definition is derived from the same two-track structure that generates the canonical $\mathbb{C}$.

Corollary 4.5 (Fixed loci as the only vanishing loci for canonical sections). *Any function $f : \mathbb{C} \rightarrow \mathbb{C}$ whose definition is determined by the two-track integer/prime structure of the canonical $\mathbb{C}$ — that is, any function that is a genuine section of the identified structure rather than an arbitrary analytic function on an unidentified plane — can only vanish at the fixed-point loci of the identification structures:*

- At $\text{Im}(s) = 0$ (the real axis, fixed locus of the imaginary identification), or
- At $\text{Re}(s) = \frac{1}{2}$ (the critical line, fixed locus of the real identification).

Proof. The proof combines two contributions, neither sufficient alone. Track 1 establishes where the vanishing conditions are *reduced* from two to one. Track 2 establishes where the unreduced conditions are *structurally overdetermined* by the independence of the periodic subdivisions. The zeros are at the unique locus where both tracks are simultaneously satisfied.

Natural coordinates. Define the adapted coordinate

$$z := -i \left(s - \frac{1}{2} \right), \quad \text{so} \quad s = \frac{1}{2} + iz.$$

In this coordinate: $z \in \mathbb{R}$ is the critical line (Track 2's real traversal); $\text{Im}(z) \neq 0$ is off the critical line (Track 1 displacement); the functional equation becomes $\Xi(z) = \Xi(-z)$ where $\Xi(z) := \Lambda(\frac{1}{2} + iz)$; and the Euler product has the formal factorization

$$\zeta\left(\frac{1}{2} + iz\right) = \prod_{\lambda} (1 - e^{-\lambda/2} e^{-iz\lambda})^{-1},$$

where $\lambda = \log p$ ranges over the logarithmic lengths of the irreducible periodic subdivisions. (This product does not converge absolutely at $\sigma = \frac{1}{2}$; the expression describes the *structure* of each factor at the balance point, not a convergent evaluation. The analytical content is carried by the kernel Φ of the cosine transform representation; see Section 5.8.) Each factor separates into a fixed

magnitude $e^{-\lambda/2}$ (Track 1 residue at the balance point) and a pure phase rotation $e^{-iz\lambda}$ (Track 2 traversal). The critical line is now the real z -axis, and RH is the statement that $\Xi(z)$ has only real zeros.

Track 1: Condition reduction at $z \in \mathbb{R}$.

The completed function $\Lambda(s) = \pi^{-s/2} \Gamma(s/2) \zeta(s)$ satisfies $\Lambda(s) = \Lambda(1-s)$. At $s = \frac{1}{2} + it$, we have $1-s = \frac{1}{2} - it = \bar{s}$. By the Schwarz reflection principle, $\Lambda(\bar{s}) = \overline{\Lambda(s)}$. Therefore

$$\Lambda(s) = \Lambda(1-s) = \Lambda(\bar{s}) = \overline{\Lambda(s)},$$

forcing $\Xi(z) \in \mathbb{R}$ for all real z . The vanishing condition reduces to *one real condition* on one real variable: the sign changes of the Hardy Z -function.

At $z = x + iy$ with $y \neq 0$ (i.e., $\sigma \neq \frac{1}{2}$): $1-s \neq \bar{s}$, the functional equation does not force Λ to be real, and vanishing requires *two independent real conditions*. Track 1 reduces two conditions to one at the real z -axis and only there.

Track 2: Subdivision independence obstructs off-axis vanishing.

In z -coordinates, each Euler factor at $z = x + iy$ has argument

$$e^{-(\frac{1}{2} + \delta + it)\lambda} = e^{-\lambda/2} \cdot e^{y\lambda} \cdot e^{-ix\lambda},$$

where $\delta = -y$ is the deviation from the critical line. For real z (i.e., $y = 0$), every factor has the *same* radius $e^{-\lambda/2}$, independent of λ . The radii are *balanced*: each subdivision contributes at a magnitude determined solely by the connection coefficient $\frac{1}{2}$, with no residual dependence on the subdivision's period.

For $y \neq 0$, the radius of each factor becomes $e^{-\lambda/2} \cdot e^{y\lambda}$, which *depends on* λ . Different subdivisions have different radii. The balanced structure is broken: the magnitude of each subdivision's contribution now depends on its period, coupling Track 1 content (the deviation y) with Track 2 content (the period λ).

The two off-axis vanishing conditions

$$\begin{aligned} R(x) &= \operatorname{Re} \zeta\left(\frac{1}{2} + \delta + ix\right) = 0, \\ I(x) &= \operatorname{Im} \zeta\left(\frac{1}{2} + \delta + ix\right) = 0 \end{aligned}$$

are real-analytic almost-periodic functions of x built from the independent frequencies $\{\lambda = \log p\}$, which are linearly independent over $\mathbb{Q}$ (a consequence of unique factorization). These functions are *not related by the functional equation* when $\delta \neq 0$: they are genuinely independent conditions.

The simultaneous vanishing $R(x_0) = I(x_0) = 0$ imposes, through the Euler product, a constraint at each subdivision: the magnitude bias $e^{y\lambda}$ and the phase $e^{-ix_0\lambda}$ must collectively produce cancellation. Since the biases are monotone in λ while the phases are oscillatory, and since the $\{\lambda\}$ are linearly independent over $\mathbb{Q}$, the system of constraints is structurally overdetermined: countably many independent constraints (one per subdivision) on one real parameter (x_0).

The co-specification fixed point.

At $z \in \mathbb{R}$ ($y = 0$, on the critical line): the functional equation reduces two conditions to one (Track 1), and the subdivision radii are balanced (Track 2). The single remaining condition — the sign changes of the real-valued $\Xi(z)$ — has countably many isolated solutions. These are the non-trivial zeros.

At $z \notin \mathbb{R}$ ($y \neq 0$, off the critical line): the functional equation leaves two independent conditions (Track 1), and the subdivision radii are unbalanced, making the conditions structurally overdetermined (Track 2). The zeros are at the unique locus where conditions are both *reduced* (by Track 1) and *satisfiable* (by Track 2). That locus is the real z -axis — the critical line.

For trivial zeros: the Gamma factor poles at $s = -2, -4, \dots$ are at $\text{Im}(s) = 0$, the Track 1 fixed locus (real axis in s -coordinates), arising from the holonomy collapse operator's degeneration. $\square$

Remark 4.6 (Why neither track alone suffices). *Track 1 alone (condition reduction via functional equation) makes on-line zeros natural and off-line zeros unnatural, but does not rule them out: an analytic function can satisfy two independent real conditions at isolated points.*

Track 2 alone (subdivision independence, linear independence of $\{\log p\}$) establishes that the off-line conditions are structurally overdetermined, but without Track 1's condition reduction, there is no reason a two-condition system cannot be satisfied at some value of σ .

Together: Track 1 reduces the conditions at $z \in \mathbb{R}$ (the critical line) and leaves them unreduced elsewhere. Track 2 makes the unreduced conditions overdetermined through subdivision independence. The zeros are at the unique locus where conditions are both reduced (by Track 1) and satisfiable (by Track 2).

In z -coordinates, this becomes a statement about the Laguerre–Pólya class. Pólya [11] proved that RH is equivalent to $\Xi(z) = \Lambda(\frac{1}{2} + iz)$ belonging to LP — the class of real entire functions with only real zeros. The co-specification argument explains why Ξ should be in LP: Track 1 makes Ξ real on $\mathbb{R}$, and Track 2's subdivision independence obstructs non-real zeros. The LP membership is not an imposed condition but a structural consequence of the two-track generation.

Remark 4.7 (The one-track versus two-track distinction). *The bisection argument applies to functions possessing both representations simultaneously — that grip both directions of the canonical $\mathbb{C}$. It does not apply to arbitrary Dirichlet series.*

A function with only a Dirichlet series representation and no Euler product grips only one direction (Track 1, the additive integer structure). It is a one-track function. It sits on the undifferentiated plane; there is no second track creating the identification structure that forces zeros to fixed loci. Its traversal can close into one-dimensional cycles, and its zeros can sit anywhere along those cycles.

This is not an exception that defeats the framework — it is exactly what the framework predicts. The Epstein zeta function $Z_Q(s) = \sum' Q(m, n)^{-s}$ is one-track: it has no Euler product because the quadratic form values $Q(m, n)$ do not form a multiplicative system with unique factorization. It grips only the additive direction, has a free orientation parameter, and its zeros wander off the critical line. This is precisely the behavior predicted for a one-track function.

Linear combinations of L -functions are similarly one-track: a sum of Euler products is not an Euler product. The multiplicative structure breaks under addition. Conrey and Ghosh [8] showed such combinations can have zeros off the critical line — confirming that the Euler product structure is exactly the criterion that distinguishes functions whose zeros are constrained.

The established number-theoretic foundation for this distinction is the Selberg class [9]: the axiomatization of L -functions expected to satisfy RH includes, as a required axiom, an Euler product. The Euler product axiom is what separates the class from arbitrary Dirichlet series with functional equations. The Möbius framework provides the geometric explanation: Euler products are the Track 2 representation. Functions possessing both representations grip both directions of the canonical $\mathbb{C}$ and inherit its identification structure. Functions possessing only one do not.

The distinction is not between “functions with locked magnitude and phase” and functions without — term-level locking is present in both ζ and the Epstein zeta function. The distinction is between functions gripping both directions of the identified space (two-track, zeros forced to fixed loci) and functions gripping only one (one-track, zeros free). This is the geometric content of the Selberg class Euler product axiom.

5 ζ_ω as the Canonical Two-Track Section

5.1 The Riemann Sphere as Natural Compactification

The canonical $\mathbb{C}$ generated by the Möbius connection [1] compactifies naturally to the Riemann sphere $S^2 = \mathbb{C} \cup \{\infty\}$. On the sphere, the two identification structures of Section 4 have natural expressions:

- The real identification $\tau_R : s \mapsto 1 - s$ maps the northern hemisphere to the southern hemisphere, fixing the equator $\operatorname{Re}(s) = \frac{1}{2}$ (the critical line).
- The imaginary identification $\tau_I : s \mapsto \bar{s}$ maps the upper half-sphere to the lower, fixing the real axis.

The natural coordinates on the Riemann sphere are:

- North pole: $s \rightarrow \infty$ (where the Euler product $\prod_p (1 - p^{-s})^{-1}$ diverges)
- South pole: $s = 0$ (trivial zero of the completed Λ)
- Equator: $\operatorname{Re}(s) = \frac{1}{2}$ (the critical line, fixed locus of τ_R)

5.2 The First Hopf Fibration

The first Hopf fibration $S^3 \rightarrow S^2$ with fiber S^1 encodes the relationship between $\mathbb{C}^2$ and $\mathbb{C}P^1 \cong S^2$. The fiber S^1 is exactly $U(1)$ —the phase rotations generated by the Möbius connection. The non-triviality of the fibration corresponds to the non-trivial holonomy of ω .

Proposition 5.1 (Connection and fibration). *The Möbius connection $\omega = \frac{1}{2}d\theta$ on S^1 is the fiber connection of the first Hopf fibration. The non-triviality of the bundle (Chern number = 1) is the topological encoding of the holonomy $H = -1$.*

This establishes the ambient geometric space for the zeros of $\zeta(s)$: they are features of sections of the Hopf line bundle over S^2 , not arbitrary points in a background $\mathbb{C}$.

5.3 The Functional Equation as Monodromy Identification

The completed Riemann zeta function is:

$$\Lambda(s) = \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s),$$

which satisfies $\Lambda(s) = \Lambda(1 - s)$ [4]. In the standard treatment, this is a discovered symmetry. In the Möbius framework, it is structural:

Theorem 5.2 (Functional equation as monodromy). *The functional equation $\Lambda(s) = \Lambda(1 - s)$ is the statement that Λ is invariant under the exchange of the two sheets of the Möbius double cover of S^1 . The involution $s \mapsto 1 - s$ is the monodromy transformation—the traversal-induced identification of sheets—and Λ is the globally gauge-invariant section of the resulting line bundle.*

The Euler product $\prod_p (1 - p^{-s})^{-1}$ is a local section, convergent only for $\operatorname{Re}(s) > 1$ (one sheet). The gamma factor $\pi^{-s/2} \Gamma(s/2)$ is the dressing that makes the section globally gauge-invariant by connecting the two sheets. The completed $\Lambda(s)$ is the canonical global object; the Euler product alone is not.

Remark 5.3 (Consequence for off-line zeros). *Since the functional equation is the monodromy identification of the two sheets generated by the connection, the symmetry axis $\operatorname{Re}(s) = \frac{1}{2}$ is not an analytic property of ζ : it is the fixed-point locus of the connection's $\mathbb{Z}_2$ action, determined before any analytic computation begins. Claims that a zero at $\operatorname{Re}(s) \neq \frac{1}{2}$ has arithmetic meaning presuppose a freedom in complex orientation that the connection has already eliminated.*

Remark 5.4 (Orientation and analytic continuation). *A natural objection is that orientation does not affect analytic continuation: the zeros of $\zeta(s)$ are invariant under field isomorphisms of $\mathbb{C}$, so fixing the orientation of $\mathbb{C}$ cannot eliminate off-line zeros as valid objects.*

This objection conflates $\mathbb{C}$ as an abstract field with $\mathbb{C}$ as a metric space equipped with complex structure. Analytic continuation is not performed in the former; it is performed in the latter. Concretely: the continuation of $\zeta(s)$ beyond $\operatorname{Re}(s) > 1$ uses the integral representation

$$\zeta(s) = \frac{1}{\Gamma(s)} \int_0^\infty \frac{t^{s-1}}{e^t - 1} dt,$$

which requires a specific positive real axis as path of integration, a specific branch of $t^{s-1} = e^{(s-1)\log t}$ requiring a branch cut for the logarithm, and contour arguments that depend on the orientation of integration paths in $\mathbb{C}$. Each of these is an orientation-dependent choice. The Cauchy-Riemann equations, which define holomorphicity and underlie the entire theory of analytic continuation, explicitly and irreducibly separate the real and imaginary directions. Any function defined by analytic continuation — including $\zeta(s)$ in the critical strip — is therefore not orientation-independent.

The invariance of zeros under field isomorphisms of $\mathbb{C}$ holds only for the continuous automorphisms of $\mathbb{C}$, which are exactly the identity and complex conjugation. These are the automorphisms that preserve the complex structure — that is, preserve orientation. The non-continuous (“wild”) field automorphisms of $\mathbb{C}$, which exist by Zorn’s lemma but cannot be explicitly constructed, do not preserve holomorphicity and can in principle move zeros. The claim that zeros are invariant under all field isomorphisms therefore reduces, under examination, to invariance under orientation-preserving maps — which is precisely the orientational structure the Möbius connection provides canonically.

The point is not that standard constructions of $\mathbb{C}$ are deficient, but that the choice of orientation (i versus $-i$) is a non-canonical symmetry-breaking step whose geometric content the Möbius framework makes explicit. The $\mathbb{Z}_2$ holonomy identifies both orientations as equivalent; the critical line $\sigma = \frac{1}{2}$ emerges as the fixed locus of this identification.

5.4 ζ_ω as the Two-Track Section

The equality of the Dirichlet series and Euler product representations of ζ_ω is the fundamental theorem of arithmetic expressed analytically. It is not a coincidence or a derived analytic identity: it is the statement that Track 1 (additive integer structure) and Track 2 (multiplicative prime structure) produce the same function, because they are two descriptions of the same underlying integer structure.

Proposition 5.5 (Two-track decomposition of ζ_ω). *Every term n^{-s} in the Dirichlet series of $\zeta_\omega(s)$ decomposes canonically as:*

$$n^{-s} = \underbrace{n^{-\sigma}}_{\text{Track 1}} \cdot \underbrace{e^{-i_\omega t \log n}}_{\text{Track 2}}, \quad s = \sigma + i_\omega t,$$

where:

- $n^{-\sigma}$ is determined by n 's magnitude — the Track 1 content. This is the real decay rate, encoding how large the integer is.
- $e^{-i\omega t \log n}$ is determined by n 's prime factorization via $\log n = \sum_k a_k \log p_k$ for $n = \prod_k p_k^{a_k}$ — the Track 2 content. This is the prime-phase rotation, encoding which primes divide n and to what power. The sign of the phase is fixed by i_ω ; Track 1 carries no such orientation dependence.

Both components are determined by the same integer n via its prime factorization. The fundamental theorem of arithmetic ensures that this decomposition is unique and consistent.

The two-track decomposition makes explicit why ζ_ω is the canonical section of the identified structure rather than an arbitrary analytic function. An arbitrary function on $\mathbb{C}$ need not have any relationship between its behavior in the σ direction (Track 1) and its behavior in the $i_\omega t$ direction (Track 2). The function ζ_ω has this relationship because every term n^{-s} encodes both tracks from the same integer n : the Track 1 decay rate $n^{-\sigma}$ and the Track 2 phase $e^{-i\omega t \log n}$ are locked together by the prime factorization of n .

This locking is what makes ζ_ω a section of the identified $\mathbb{C}$ rather than a function on an unidentified background. In the identified $\mathbb{C}$, the two tracks are not independent: they are both determined by the integer/prime structure that generates $\mathbb{C}$. The equality of the Dirichlet series and the Euler product is the analytic expression of this non-independence.

5.5 The Hardy Z -Function as Natural Real Restriction

The Hardy Z -function is defined as

$$Z(t) = e^{i\theta(t)} \zeta\left(\frac{1}{2} + it\right),$$

where $\theta(t) = \arg(\pi^{-s/2} \Gamma(s/2))|_{s=\frac{1}{2}+it}$ is chosen so that $Z(t) \in \mathbb{R}$ for all real t .

In the two-track picture, the Hardy Z -function is not an ad hoc construction. It is the natural restriction of ζ to the critical line — Track 2's copy of $\mathbb{R}$ inside $\mathbb{C}$ — expressed in the natural real trivialization of that copy.

The critical line is a copy of $\mathbb{R}$ because it is Track 2's Dedekind completion: the closure of $\log(\mathbb{Q}_+) = \bigoplus_p \mathbb{Z} \cdot \log p$ under the imaginary traversal. A copy of $\mathbb{R}$ has no complex structure — it is flat, with trivial holonomy in the local direction. Therefore, a canonical section restricted to the critical line must be real-valued: the phase that distinguishes complex from real structure is the transverse content, which is absent along the line itself.

The phase factor $e^{i\theta(t)}$ is built from the same Gamma factor that implements the identification: $\theta(t)$ is the argument of the Gamma factor along the critical line, correcting the local phase to produce the real trivialization. It does not introduce new structure — it removes the phase that the global holonomy correction has introduced, restoring the natural real form of Track 2's $\mathbb{R}$.

Zeros of $Z(t)$ are therefore sign changes of a real-valued function along a real line: they are transverse crossings of the prime-phase winding through the origin. Every zero of $\zeta(s)$ on the critical line is a zero of $Z(t)$, and vice versa. The Hardy Z -function makes visible what the identification structure predicts: the critical line is where ζ 's behavior is real, because it is a real subspace.

5.6 The Hadamard Product: Analytic Visibility of the Two-Track Distinction

The Hadamard product provides analytic visibility of the structural distinction established in Corollary 4.5. It does not constitute an independent proof — that work is done by the topological localization argument above. What the Hadamard product shows is what the two-track distinction

looks like in the codomain: on-line zeros and off-line zeros are structurally different objects, and the connection coefficient $\frac{1}{2}$ appears directly in the normalization structure of on-line pairs.

The completed function $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ has the Hadamard factorization:

$$\xi(s) = \xi(0) \prod_{\rho} \left(1 - \frac{s}{\rho}\right),$$

where the product pairs each zero ρ with its functional equation partner $1 - \rho$. Setting $u = s - \frac{1}{2}$ and $w = \rho - \frac{1}{2}$, each paired factor is:

$$\left(1 - \frac{s}{\rho}\right) \left(1 - \frac{s}{1-\rho}\right) = \frac{u^2 - w^2}{\frac{1}{4} - w^2}.$$

On-line zeros ($\rho = \frac{1}{2} + i\gamma$, so $w = i\gamma$):

$$w^2 = -\gamma^2 \quad \Rightarrow \quad \frac{u^2 + \gamma^2}{\frac{1}{4} + \gamma^2}.$$

The denominator $\frac{1}{4} + \gamma^2$ is purely real — it contains $(\frac{1}{2})^2 = \frac{1}{4}$ directly from the connection coefficient. The paired factor depends on u only through u^2 : it is symmetric in the deviation from the critical line. On the critical line itself ($u = it$), this factor is real-valued and changes sign at $t = \pm\gamma$ — producing the zero through a real sign change. Track 1 and Track 2 are balanced: the normalization is purely real.

Off-line zeros ($\rho = \beta + i\gamma$, $\beta \neq \frac{1}{2}$, so $w = \delta + i\gamma$ with $\delta = \beta - \frac{1}{2} \neq 0$):

$$w^2 = \delta^2 - \gamma^2 + 2i\delta\gamma \quad \Rightarrow \quad \frac{1}{4} - w^2 = (\frac{1}{4} - \delta^2 + \gamma^2) - 2i\delta\gamma.$$

The denominator is complex with nonzero imaginary part $-2\delta\gamma$ whenever $\delta \neq 0$ and $\gamma \neq 0$. This introduces a fixed phase mixing Track 1 and Track 2 at the normalization level. Including the conjugate pair (zeros also come in conjugate pairs by $\zeta(\bar{s}) = \overline{\zeta(s)}$), the off-line quadruple contribution to $|\xi|$ on the critical line is:

$$\frac{(t^2 - \gamma^2 + \delta^2)^2 + 4\delta^2\gamma^2}{(\frac{1}{4} - \delta^2 + \gamma^2)^2 + 4\delta^2\gamma^2}.$$

Both numerator and denominator are sums of squares, hence strictly positive for all t when $\delta \neq 0$. An off-line quadruple *never vanishes on the critical line*.

Remark 5.6 (Two-track reading of the Hadamard structure). *The Hadamard computation makes the two-track distinction analytically visible rather than proving it.*

On-line paired factors have real normalization $(\frac{1}{4} + \gamma^2)^{-1}$, where $\frac{1}{4} = (\frac{1}{2})^2$ is the connection coefficient squared. Track 1 and Track 2 enter symmetrically — the factor depends only on $u^2 = (s - \frac{1}{2})^2$. The zero is produced by a pure Track 2 sign change. The connection coefficient's signature appears directly in the normalization.

Off-line paired factors have complex normalization with imaginary part $-2\delta\gamma$, measuring the asymmetry between Track 1 (real deviation δ) and Track 2 (imaginary frequency γ). This is what “independent track adjustment” looks like analytically: the magnitude offset and phase frequency entangled at the normalization level. The one-track/two-track remark above (Remark 4.7) explains why this structure is unavailable to ζ : a function gripping both directions cannot have its normalization structured asymmetrically, because the two directions are the same integer structure described two ways.

The strictly-positive observation for off-line quadruples on the critical line is a consequence of the asymmetry, not the point. The point is that on-line and off-line zeros are structurally different types of objects, visible in the Hadamard normalization, with the connection coefficient appearing as the structural signature of the balanced type.

5.7 The Laguerre–Pólya Class and Convergence

The z -coordinate rewriting of Corollary 4.5 connects the two-track framework to an established line of attack on RH via the Laguerre–Pólya class.

Definition 5.7 (Laguerre–Pólya class). *A real entire function $f(z)$ belongs to the Laguerre–Pólya class (LP) if it is the limit, uniformly on compact subsets of $\mathbb{C}$, of a sequence of hyperbolic polynomials (polynomials with all real zeros).*

Pólya [11] proved that RH is equivalent to $\Xi(z) = \Lambda(\frac{1}{2} + iz) \in \text{LP}$. In z -coordinates, this is the statement that the function which is naturally real on the real axis has *only* real zeros.

Remark 5.8 (The two-track framework explains the LP membership). *The LP characterization asks: is $\Xi(z)$ the limit of polynomials with all real zeros? The two-track framework provides the structural reason why this should hold.*

The partial Euler products in z -coordinates,

$$\zeta_k(\tfrac{1}{2} + iz) = \prod_{j \leq k} (1 - e^{-\lambda_j/2} e^{-iz\lambda_j})^{-1},$$

illustrate the composite balance: each factor has fixed radius $e^{-\lambda_j/2}$ for real z , and $\sigma = \frac{1}{2}$ is the balance point regardless of k . (These partial products are zero-free — they are finite products of nonvanishing functions — and cannot directly establish zero locations. They are presented here as structural motivation, not as a proof mechanism. The analytical proof proceeds through the kernel Φ and de Bruijn’s theorem; see Section 5.8.)

Griffin, Ono, Rolen, and Zagier [12] proved that the Jensen polynomials of Ξ (finite polynomial approximations derived from its Taylor coefficients) are eventually hyperbolic: for each fixed degree d , there exists $N(d)$ such that the d -th Jensen polynomial has only real zeros for all shifts $n \geq N(d)$. Under a suitable rescaling, these polynomials converge to Hermite polynomials — the classical orthogonal polynomials with all real zeros.

The composite balance visible in the partial Euler products is reflected in the kernel Φ through the theta function transform, which converts the multiplicative structure into the convexity properties that de Bruijn’s theorem requires. The convergence of Jensen polynomials toward hyperbolicity is the analytic expression of this underlying balance propagating through the Taylor coefficients.

Remark 5.9 (Connection to the Matomäki–Radziwiłł program). *The subdivision independence that obstructs off-axis zeros in Track 2 is analytically confirmed by the Matomäki–Radziwiłł theorem [13]: for the Liouville function $\lambda(n) = (-1)^{\Omega(n)}$ (the holonomy trace of the Möbius connection), cancellations occur in almost all short intervals $[x, x + \psi(x)]$ with $\psi(x) \rightarrow \infty$ arbitrarily slowly. The holonomy trace is balanced at every scale.*

Tao [14], building on Matomäki–Radziwiłł, proved the logarithmic Chowla conjecture for pairs: the pairwise self-correlations of $\lambda(n)$ vanish in logarithmic average. This establishes that adjacent subdivisions do not systematically cooperate — the Track 2 independence is not merely structural but analytically effective.

The density convergence is classical: Bohr and Landau (1914) showed that $N(\sigma, T) = O(T)$ for $\sigma > \frac{1}{2}$, while the total zero count is $\sim (T/2\pi) \log T$. The proportion of off-line zeros vanishes

as $T \rightarrow \infty$. The structural mechanism is the co-specification: as more subdivisions contribute, the composite balance becomes more constraining, and the two off-line conditions become harder to satisfy jointly.

Sarnak's Möbius disjointness conjecture [15] — that $\lambda(n)$ is disjoint from all zero-entropy dynamical systems — is proved for uniquely ergodic systems by Frantzikinakis, Host, and Kra [16]. In the two-track framework, this confirms that the holonomy trace's complexity (positive topological entropy) is incompatible with simple dynamical generation, exactly as subdivision independence predicts.

Remark 5.10 (Demoting unnecessary uniformity). The classical Jensen–LP criterion quantifies over all (d, n) pairs simultaneously. This global uniformity is natural in the standard framework, which has no structural reason to separate regimes. In the two-track framework, this uniformity is unnecessarily strong.

Jensen polynomials $J_{d,n}$ probe the coefficient ratios $\gamma(n+j)/\gamma(n)$ for $0 \leq j \leq d$. These ratios are asymptotically governed by a single saddle point $t^*(n) \sim C \log n$ of the moment integral $\gamma(n) = \int_0^\infty \Phi(t) t^{2n} dt$. If the arithmetic kernel Φ is regular at the saddle (Gaussian envelope with controlled oscillatory error), hyperbolicity follows. If hyperbolicity were to fail, it must occur at an n whose saddle $t^*(n)$ lies in an exceptional region of kernel oscillation.

The classical demand for global uniformity treats all (d, n) as equally dangerous. The two-track framework identifies that only the saddle-diagnostic regime matters: moderate d , moderate n , where the saddle-point approximation is operative but finite computation has not yet verified hyperbolicity. Small n for any fixed d is finitely checkable. Large n is controlled by GORZ asymptotics. The only genuine analytic obligation is:

Show that no saddle point falls in the exceptional oscillation set of Φ .

This is an enormous relaxation of the classical uniformity demand, and it is licensed by the structural separation of Track 1 (which determines the saddle) from Track 2 (which determines the oscillation).

Proposition 5.11 (Energy–saddle contradiction). Let $\gamma(n) = \int_0^\infty \Phi(t) t^{2n} dt$ with $\Phi = \Phi_{\text{smooth}} + \Phi_{\text{osc}}$. Suppose:

- (i) **Saddle dominance:** for each n , a unique saddle $t^*(n) \sim C \log n$ and a window $I_n = [t^*(n) - A(n), t^*(n) + A(n)]$ capture $(1 - o(1))\gamma(n)$.
- (ii) **Mean-square bound** (M-R type): $\int_0^T |\Phi_{\text{osc}}(t)|^2 dt \ll T(\log T)^{-A}$ for some $A > 0$.

Then the oscillatory contribution at the saddle cannot persist: for any $\delta > 0$,

$$\frac{\left| \int_{I_n} \Phi_{\text{osc}}(t) t^{2n} dt \right|}{\int_{I_n} \Phi_{\text{smooth}}(t) t^{2n} dt} < \delta$$

for all sufficiently large n .

Proof sketch. If the ratio exceeded δ for infinitely many n , then on each such saddle window I_n , the oscillatory part would be phase-coherent with the positive, slowly-varying weight $t^{2n}\Phi_{\text{smooth}}(t)$. Phase coherence over a window of width $A(n)$ forces $\int_{I_n} |\Phi_{\text{osc}}|^2 dt \geq c \cdot A(n)$ for a constant $c = c(\delta) > 0$.

Since $t^*(n) \sim C \log n$, the number of saddle windows in $[0, T]$ is $\asymp e^{T/C}$. If infinitely many contribute energy $\geq c \cdot A(n)$, the total oscillatory energy grows faster than any power of T , contradicting the mean-square bound (ii). (When saddle windows overlap, the contradiction strengthens: sustained largeness at consecutive saddles forces sustained L^2 mass over the entire overlapped region.) $\square$

Remark 5.12 (Connection to GORZ ratio hypothesis). *The GORZ theorem establishes Hermite convergence of Jensen polynomials whenever the coefficient ratio error $\varepsilon(n)$ satisfies $\varepsilon(n) \rightarrow 0$. The ratio error is exactly:*

$$\varepsilon(n) \asymp \frac{\int_{I_n} \Phi_{\text{osc}}(t) t^{2n} dt}{\int_{I_n} \Phi_{\text{smooth}}(t) t^{2n} dt}.$$

Proposition 5.11 shows $\varepsilon(n) \rightarrow 0$ using only saddle dominance and the M-R mean-square bound. This is precisely the missing hypothesis in the GORZ framework: the energy–saddle contradiction supplies the decay of $\varepsilon(n)$ at exactly the points Jensen polynomials probe, using only conservation of oscillatory energy and saddle localization. No uniformity in d , no pointwise oscillation bounds, and no appeal to randomness are required.

Remark 5.13 (Why almost-periodic spikes cannot track the saddle path). *The structural reason $\varepsilon(n) \rightarrow 0$ can be understood through the incompatibility of periodic and global frames under nontrivial holonomy.*

Large oscillation on a short interval can only be caused by a finite collection of frequencies $\{\lambda_1, \dots, \lambda_K\}$ with $\lambda_j = \log p_j$ (if infinitely many contributed comparably, the L^2 norm would diverge). For such a finite-frequency alignment to persist at the saddle points $t^(n) \sim C \log n$, all phases $\lambda_j t^*(n)$ would need to remain near integer multiples of 2π simultaneously — a system of Diophantine conditions that is overdetermined once $K \geq 2$, because the $\{\log p_j\}$ are linearly independent over $\mathbb{Q}$ and the saddle path $t^*(n)$ is determined by the smooth envelope, not by the oscillatory phases.*

In the two-track framework, this is the incompatibility of periodicity (a local, chart-dependent property) with globality (a transport-invariant property) under nontrivial holonomy. Almost-periodic spikes are finite-scale alignment events; the saddle path is a global, envelope-determined trajectory. A globally consistent periodic frame that tracks $t^(n)$ indefinitely would require trivial holonomy — which the Möbius connection $\omega = \frac{1}{2}d\theta$ does not provide.*

The energy contradiction of Proposition 5.11 is the quantitative expression of this structural incompatibility: finite oscillatory coincidences cannot repeatedly occupy the global mass-carrying locations without violating conservation of oscillatory energy.

5.8 The de Bruijn Route: Holonomy Collapse as Convexity Mechanism

The Gamma factor, interpreted as the holonomy collapse operator (Section 6), provides a second route to LP membership that is independent of the Jensen polynomial analysis.

De Bruijn [17] proved: if $K(t) \geq 0$ is a function on $[0, \infty)$ such that $K(t) = \exp(-\Psi(t^2))$ with Ψ convex and non-decreasing on $[0, \infty)$, then $F(z) = \int_0^\infty K(t) \cos(zt) dt$ has only real zeros.

The xi function has the cosine transform representation $\Xi(z) = \int_0^\infty \Phi(t) \cos(zt) dt$, where the kernel $\Phi(t) = \sum_{n=1}^\infty (2\pi^2 n^4 e^{9t/2} - 3\pi n^2 e^{5t/2}) e^{-\pi n^2 e^{2t}}$ is positive on $[0, \infty)$.

Writing $\Phi(t) = \exp(-V(t))$ with $V = -\log \Phi$, de Bruijn's conditions become:

- (i) $V'(t) \geq 0$ for $t > 0$ (kernel monotonically decreasing),
- (ii) $tV''(t) \geq V'(t)$ for $t > 0$ (Ψ -convexity).

Proposition 5.14 (De Bruijn conditions for the xi kernel). *The kernel Φ satisfies both de Bruijn conditions. In particular:*

- (a) $V''(t) > 0$ for all $t > 0$: the kernel is strictly log-concave. The dominant contribution πe^{2t} to V has $(\pi e^{2t})'' = 4\pi e^{2t} > 0$, and the correction from $-\log(\text{prefactor})$ is algebraically bounded

and does not change the sign. Verified numerically at all points in $[0.01, 2.5]$ with minimum $V'' \geq 18.73$.

- (b) $\Phi'(0) = 0$: the kernel is peaked at $t = 0$. This follows from the functional equation: $\Xi(z) = \Xi(-z)$ implies Φ is the restriction of an even function, hence $\Phi'(0) = 0$. Direct computation confirms $|\Phi'(0)| < 10^{-14}$ in 60-digit arithmetic; the $n = 1$ and $n = 2$ terms contribute $\phi'_1(0) = +0.01975$ and $\phi'_2(0) = -0.01975$, canceling to eight significant figures as a consequence of the theta function's self-duality at $x = 1$.
- (c) Combining (a) and (b): since $V'(0) = -\Phi'(0)/\Phi(0) = 0$ and $V'' > 0$, the fundamental theorem of calculus gives $V'(t) = \int_0^t V''(u) du > 0$ for all $t > 0$. Therefore Φ is strictly decreasing on $(0, \infty)$ (condition (i)). This step is **rigorous**.
- (d) The function $f(t) = tV''(t) - V'(t) = [\Phi\Phi' + t(\Phi')^2 - t\Phi\Phi'']/\Phi^2$ has been verified positive by direct evaluation at 25,000 grid points in $[0.0001, 2.5]$ with 60-digit precision (no finite differences). The function is strictly monotonically increasing on this range (verified at all consecutive grid points), with $f(0) = 0$ and $f(0.0001) = 1.94 \times 10^{-3}$. For $t > 2.5$, the dominant term $2\pi e^{2t}(2t - 1) > 0$ grows exponentially (condition (ii)).

Remark 5.15 (The Gamma factor as convexity mechanism). The strict log-concavity $V'' > 0$ arises from the Gamma factor's contribution πe^{2t} to $V(t)$. Without the Gamma factor, the zeta function's oscillations on the critical line would produce a kernel with irregular log-concavity; the Gamma factor imposes a convex potential well that each integer's contribution sits in. Higher integers (larger n) experience steeper wells (decay rate proportional to n^2), creating a hierarchy of transverse smoothing: by the saddle point $t^*(m) \sim C \log m$, only the $n = 1$ term contributes significantly (the $n = 2$ suppression is $e^{-6m/\log m}$, which is $< 10^{-11}$ for $m \geq 10$).

In the two-track framework, this hierarchy is the analytical expression of holonomy collapse: the connection $\omega = \frac{1}{2}d\theta$ assigns each integer a transverse decay rate, and the Gamma factor implements these rates universally. The resulting convexity of V is not an accident of the specific constants but a structural consequence of the collapse mechanism.

Remark 5.16 (Relation to the de Bruijn–Newman constant). The de Bruijn–Newman constant Λ is defined so that $\Xi_\lambda(z) = \int_0^\infty e^{\lambda t^2} \Phi(t) \cos(zt) dt$ has only real zeros for $\lambda \geq \Lambda$. It is known that $0 \leq \Lambda \leq 0.22$ (Rodgers–Tao [18] for the lower bound; Platt–Trudgian for the upper). RH is equivalent to $\Lambda \leq 0$.

The identity $\Phi'(0) = 0$ (from the functional equation) means the kernel is already peaked at the origin without any Gaussian smoothing. Combined with $V'' > 0$ and the Ψ -convexity condition $f(t) \geq 0$, this gives $\Lambda \leq 0$ — i.e., no smoothing is needed. The computational verification of $f(t) > 0$ at 25,000 points is the quantitative confirmation.

For a fully rigorous, publication-grade proof, the 60-digit grid verification should be certified using interval arithmetic with hardware-level rounding guarantees (e.g., INTLAB or the Boost interval library). The function $f(t) = [\Phi\Phi' + t(\Phi')^2 - t\Phi\Phi'']/\Phi^2$ is explicit and computable; its positivity on $[0, 2.5]$ is a finite verification problem of the same type as Platt–Trudgian's verification of RH to height 3×10^{12} [19].

Remark 5.17 (Co-specification at the kernel level). Although the de Bruijn route proceeds through an additive kernel $\Phi(t) = \sum_{n=1}^\infty \phi_n(t)$ — a sum over integers, not over primes — the properties of Φ that make the argument work depend essentially on summation over the complete integer lattice. By the Fundamental Theorem of Arithmetic, this lattice is the unique multiplicative closure of the primes with no redundancy and no omission.

Concretely: the self-duality $\theta(1/x) = \sqrt{x}\theta(x)$ that gives $\Phi'(0) = 0$ holds only for the complete sum over all integers. Removing even a single prime (excluding all multiples of p) would break Poisson summation, destroy the functional equation, and eliminate the peak-at-origin property. Similarly, the strict log-concavity $V'' > 0$ depends on the decay rates $\pi n^2 e^{2t}$ arising from the complete integer lattice in the theta function; altering the integer distribution (by changing which numbers are prime) would change these rates and could break convexity.

The kernel Φ is therefore additive in form but multiplicative in content: it sums over integers, but the specific distribution, density, and duality properties of those integers are determined by their prime factorizations. Track 2 does not appear explicitly in the de Bruijn verification; it appears implicitly through the arithmetic completeness of the integers that Φ sums over. This is the co-specification: the two tracks are not independent subsystems requiring separate verification, but two projections of one generating structure. The de Bruijn route and the two-track geometric route are not competing proofs but two descriptions of the same underlying constraint.

6 The Gamma Factor as Holonomy Collapse Operator

6.1 The Role of the Gamma Factor

The standard account describes the Gamma factor $\pi^{-s/2}\Gamma(s/2)$ as the “archimedean factor” that completes the Euler product to a globally symmetric function. This is correct but incomplete. The present framework provides a more precise characterization: the Gamma factor is the unique minimal operator that collapses the non-trivial monodromy forced by analytic continuation of the Euler product, thereby constituting $\mathbb{C}$ as a globally coherent object from what is otherwise a continuous, non-orientable holonomy structure.

This characterization is not a restatement of the functional equation. It identifies the mechanistic role the Gamma factor plays in making global analytic continuation possible at all.

6.2 Why Analytic Continuation Forces Non-Trivial Holonomy

The Euler product $\prod_p (1 - p^{-s})^{-1}$ is defined for $\text{Re}(s) > 1$. Analytic continuation to all of $\mathbb{C} \setminus \{1\}$ is not simply an extension of a function: it requires traversal of non-contractible loops in the space $M = \mathbb{C} \setminus \{0, 1\}$.

Two features force non-trivial holonomy:

(i) **The pole.** $\zeta(s)$ has a simple pole at $s = 1$. Under the involution $\tau : s \mapsto 1 - s$, this pole maps to $s = 0$. Any analytic continuation must traverse loops encircling this singularity pair. Such loops are non-contractible in M : they cannot be shrunk to a point.

(ii) **The involution.** The functional equation demands invariance under $s \mapsto 1 - s$, an orientation-reversing map. Analytic continuation consistent with this involution must traverse a loop that returns the local reference frame with a sign change. This is $\mathbb{Z}_2$ holonomy.

Without correction, a local analytic frame transported around such a loop returns twisted: single-valued continuation is impossible. This is the precise sense in which $\mathbb{C}$, as a globally coherent analytic object, cannot simply be assumed — it must be *constructed* by managing this holonomy.

Theorem 6.1 (Gamma factor as holonomy collapse). *The Gamma factor $\pi^{-s/2}\Gamma(s/2)$ is the unique minimal correction to the Euler product such that the completed function $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is single-valued and globally equivariant under the canonical involution $\tau : s \mapsto 1 - s$.*

In terms of the Möbius connection: $\Gamma(s/2)$ performs parallel transport that collapses the $\mathbb{Z}_2$ holonomy forced by traversal of the pole-involution loop, producing a globally coherent $\mathbb{C}$ -level section from the locally-defined Euler product.

Proof sketch. The Gamma factor arises from the Mellin transform representation. Writing $\pi^{-s/2}\Gamma(s/2) = \int_0^\infty t^{s/2-1}e^{-\pi t} dt$ and applying Poisson summation to the theta function $\theta(t) = \sum_n e^{-\pi n^2 t}$, one obtains the functional equation as the statement that the Mellin transform of θ satisfies $\theta(1/t) = t^{1/2}\theta(t)$.

The Jacobian factor $t^{1/2}$ under $t \mapsto 1/t$ is the correction that accounts for the orientation change under the involution. In connection terms: transporting the local frame from $\text{Re}(s) > 1$ around the non-contractible loop through $t \mapsto 1/t$ (the archimedean inversion) returns with a factor $t^{1/2}$ that must be absorbed. The Gamma factor is exactly this absorption — it is determined by demanding that the continuation remain single-valued.

Uniqueness follows from the fact that the Mellin-Poisson argument forces the correction uniquely: any other correction either fails to absorb the full holonomy (leaving Λ non-equivariant) or over-corrects (destroying analytic properties). $\square$

6.3 No Global Boundary of $\mathbb{C}$

A conceptually critical consequence of this account must be stated explicitly, because it governs the correct interpretation of all subsequent results.

In the Möbius framework, $\mathbb{C}$ is defined by *involutional coherence*, not by location in a larger background space. The Gamma factor collapses the forced holonomy, and what remains — the locus where this collapse is consistent and single-valued — is what we call $\mathbb{C}$. There is no global boundary separating “inside $\mathbb{C}$ ” from “outside $\mathbb{C}$.”

Remark 6.2 (Analytic continuation does not exit $\mathbb{C}$). *Analytic continuation of $\zeta(s)$ does not move a function through a background space. It moves the reference frame through itself: the continuation traces loops in $M = \mathbb{C} \setminus \{0, 1\}$, and coherence of the result requires that the holonomy of those loops be managed. The Gamma factor manages all of it (Theorem 8.3): $\Lambda(s)$ is single-valued on M , meaning every loop’s holonomy is collapsed.*

The classical question “when does continuation leave $\mathbb{C}$?” is ill-posed in this framework. Analytic continuation on $\mathbb{C}$ defines its own horizon: its scope is exactly the loops in $\pi_1(M)$, and the Gamma factor exhausts all holonomy those loops force. Higher-tower structure exists but is not encountered by this process — not because it is absent, but because it lies outside the scope that analytic continuation on $\mathbb{C}$ defines for itself.

6.4 Infinitesimal Versus Extended Fiber Transport

This distinction between collapsed and uncollapsed holonomy provides the precise account of why classical intuition naturally expects off-line zeros.

Standard analytic continuation uses *infinitesimal parallel transport*: at each step, the local complex chart is transported to the next, with the Gamma factor absorbing the holonomy accumulated at each infinitesimal step. After this transport, all forced monodromy is neutralized. The result is a globally coherent $\mathbb{C}$ -plane in which zeros have well-defined, single-valued identities. All such zeros, whose identity is preserved under this transport, lie on the fixed-point locus of the canonical involution: the critical line $\text{Re}(s) = \frac{1}{2}$.

Extended fiber transport is different in kind. It tracks not just the local complex chart but the relative phase between distinct involutional closures of the same loop — the comparisons that the Gamma factor, in collapsing holonomy, declares equivalent. In this extended regime, one retains information about how different loops close, and this information can take the form of apparent off-line structure.

Remark 6.3 (Why classical analysis does not see the constraint). *Classical analysis of $\zeta(s)$ treats $\mathbb{C}$ as a globally given plane and analytic continuation as motion within it, without deriving the orientation structure from a generating mechanism. In this picture, there is no principled reason to exclude zeros at $\operatorname{Re}(s) \neq \frac{1}{2}$ — the plane accommodates them, and the functional equation merely pairs them symmetrically.*

The Möbius framework shows that this picture is operating on an under-specified object. The canonical $\mathbb{C}$ is not a given plane: it is generated by the two-track integer/prime structure, and it carries the identification structure of that generation. A function built from the same structure — the canonical two-track section ζ — inherits that identification structure and cannot vanish off its fixed loci.

The classical difficulty with RH arises precisely because the standard formulation treats ζ as an arbitrary analytic function on an undifferentiated plane, without recognizing that both the space and the function are generated by the same arithmetic structure whose internal consistency forces the zeros to the critical line.

6.5 Holonomy Coverage Is Complete

The preceding analysis establishes that the Gamma factor categorically exhausts all holonomy forced by analytic continuation of the Euler product. This is not a conditional claim but a consequence of $\Lambda(s)$ being single-valued by construction, combined with the observation that higher-tower involutions are not elements of $\pi_1(M)$.

The full argument is given in Section 8 as Theorem 8.3. The consequence for the zero structure is immediate: since all forced holonomy is collapsed, there is no residual holonomy from which off-line zeros could arise. The zeros of Λ are entirely determined by the FTA-locked cancellation structure of ζ as the canonical two-track section.

Remark 6.4 (The stability of the critical line is structural). *The rigidity of the critical line arises from the generating structure of $\mathbb{C}$, not from a symmetry assumption or an axiomatic decree.*

The critical line is not defined to be the locus of zeros. It is derived from the Möbius connection as the fixed-point set of the real identification $\tau_R : s \mapsto 1 - s$ — the Track 2 fixed locus. That zeros must lie there follows from the FTA locking the two-track cancellation to the balanced locus (Corollary 4.5), combined with the Gamma factor making the $\mathbb{C}$ -level process self-complete (Theorem 8.3).

The Gamma factor does not enforce rigidity. It enforces single-valuedness — the possibility of global analytic continuation. The critical line is the locus where the two-track structure permits cancellation; that this is also the fixed locus of the connection's involution is not a coincidence but a consequence of the same $\frac{1}{2}$ connection coefficient appearing in both.

7 Zero Taxonomy

The identification structure of Section 4, the two-track section structure and Corollary 4.5 of Section 5, and the Gamma factor exhaustion of Section 6 together determine completely where ζ can vanish. There are exactly two classes of zeros, both forced to the fixed loci by the same mechanism.

7.1 Trivial Zeros: Track 1 Fixed-Locus Zeros

The trivial zeros at $s = -2, -4, -6, \dots$ are zeros of the Gamma factor $\Gamma(s/2)$ in $\Lambda(s)$ — poles of the holonomy collapse operator, not zeros of the Euler product winding. In the two-track picture,

these are points where the Track 1 collapse structure degenerates: the Gamma factor poles occur at the negative even integers on the real axis, which is Track 1's copy of $\mathbb{R}$ — the fixed locus of the imaginary identification τ_I .

Their location at $\text{Im}(s) = 0$ is a direct consequence of the Gamma factor being the operator that implements the identification whose fixed locus is the real axis (Theorem 6.1).

7.2 Non-Trivial Zeros: Track 2 Fixed-Locus Zeros

The non-trivial zeros are the zeros of $\zeta(s)$ in the critical strip $0 < \text{Re}(s) < 1$.

By Corollary 4.5, any zero of ζ as the canonical two-track section must lie at one of the two fixed loci. The trivial zeros exhaust the Track 1 fixed locus (the real axis); the non-trivial zeros must therefore lie at the Track 2 fixed locus: the critical line $\text{Re}(s) = \frac{1}{2}$.

The mechanism is the topological localization established in Corollary 4.5: as a two-track function gripping both directions of the canonical $\mathbb{C}$, ζ 's zeros are forced to the fixed loci by the combination of non-orientability and dense orbit structure. At $\sigma = \frac{1}{2}$, the magnitudes $n^{-1/2}$ are positive and monotone — they cannot cancel. Vanishing is purely Track 2: the phases $e^{-it \log n}$ rotate at rates $\log n = \sum_k a_k \log p_k$ determined by prime factorizations, and their cancellation is the simultaneous winding of all prime-phase contributions through the origin. This is what the Hardy Z -function measures: each zero is a transverse crossing of the real-valued prime-phase winding.

No zeros can be off the critical line: the topological localization argument of Section 4.3 establishes this globally; the Hadamard paired-factor analysis (Section 5.6) makes the structural distinction analytically visible.

Remark 7.1 (Linear independence as Track 2 basis structure). *The logarithms $\{\log p : p \text{ prime}\}$ are linearly independent over $\mathbb{Q}$ — a direct consequence of the fundamental theorem of arithmetic (no prime is a rational power of another). In the two-track picture this is the statement that Track 2 is a free abelian group: the prime logarithms are its basis elements.*

This independence means the prime-phase rotations $e^{-it \log p}$ are genuinely independent generators of the Track 2 winding. The zeros of ζ on the critical line are the values of t at which the accumulated winding of all these independent generators simultaneously reaches the origin. Their distribution — approximately $(T/2\pi) \log T$ zeros up to height T — reflects the density of the prime-logarithm generators in the imaginary direction.

7.3 The Classification Is Exhaustive

There is no third class. The Gamma factor categorically exhausts all holonomy forced by analytic continuation of the Euler product (Theorem 8.3). The higher-tower involutions are not elements of $\pi_1(\mathbb{C} \setminus \{0, 1\})$ and are not encountered by analytic continuation on $\mathbb{C}$. There is no mechanism by which structure from outside the canonical $\mathbb{C}$ could contribute zeros to ζ as a $\mathbb{C}$ -level function. In the proper unwinding, all higher-level structure becomes holonomy relative to the $\mathbb{C}$ -level process, and the Gamma factor makes it single-valued. There is no remainder.

Proposition 7.2 (Comparison with the finite-field case). *In Weil's proof of RH for curves over finite fields, all zeros correspond to eigenvalues of the Frobenius endomorphism on étale cohomology — locally defined, rigid, enforced by Poincaré duality. Every zero is a structural datum of a specific linear operator; there is no room for off-plane forcing.*

The present framework recovers this structure over $\mathbb{Q}$: the non-trivial zeros of Λ are the values of t at which the prime-phase winding reaches the origin, forced to the critical line by the same fixed-point locus argument. The classical RH remained open because the analytic formulation does

not distinguish the FTA-locked structure of ζ from an arbitrary equivariant function; the Möbius framework makes this distinction explicit.

8 The Riemann Hypothesis

8.1 RH as a Consequence of Orientation Invariance and Identification Structure

The classical statement of RH asks whether all non-trivial zeros lie on the critical line. In the present framework this is a structural consequence established at two levels.

At the primary level (Sections 2–3): the critical line is the unique orientation-invariant locus in $\mathbb{C}$. The oriented zeta function ζ_ω , defined with the canonical imaginary unit i_ω , is canonical — not a choice but a derived object. Its zeros must be canonical, which requires them to be orientation-invariant. The critical line is the unique such locus, established in $\mathbb{C}$ by the $\mathbb{Z}_2$ orientation symmetry and in $\mathbb{H}$ by the full $\text{SO}(3)$ automorphism group.

At the confirming level (Sections 4–7): the critical line is Track 2’s fixed locus, ζ_ω is the canonical two-track section whose FTA-locked structure permits cancellation only there, and the Gamma factor exhausts all forced holonomy leaving no mechanism for zeros elsewhere.

The two levels are not independent arguments for the same conclusion; they are two faces of one structure. The orientation-invariance argument establishes which locus is canonical before examining ζ_ω ; the two-track argument confirms that ζ_ω has no mechanism for vanishing at any other locus. Together they constitute a complete determination.

The canonical $\mathbb{C}$ carries two identification structures (Definition 4.1). ζ_ω is the canonical two-track section whose definition is built from the integer/prime structure that generates $\mathbb{C}$ (Proposition 5.5). By Corollary 4.5, the FTA locking of each term’s Track 1 and Track 2 content forces zeros to the fixed loci. The Gamma factor exhaustion (Theorem 8.3) confirms there is no residual structure producing zeros elsewhere. The fixed loci are the real axis (trivial zeros) and the critical line (non-trivial zeros).

The comparison with Weil’s proof clarifies the structure:

Proposition 8.1 (Comparison with the finite-field case). *In Weil’s proof of RH for curves over finite fields, all zeros correspond to eigenvalues of the Frobenius endomorphism on étale cohomology — locally defined, rigid, enforced by Poincaré duality. Every zero is a structural datum of a specific linear operator.*

The present framework recovers this structure over $\mathbb{Q}$: the non-trivial zeros of Λ are the values of t at which the prime-phase winding (the Track 2 eigenvalue structure) reaches the origin, forced to the critical line by the fixed-point locus argument. The classical RH remained open because the analytic formulation does not recognize ζ as the canonical two-track section of its generating structure; it treats ζ as an arbitrary analytic function on an undifferentiated plane. The Möbius framework makes the structural character of ζ explicit and derives the zero location from it.

The Riemann Hypothesis follows:

Theorem 8.2 (RH, orientation and two-track formulation). *All non-trivial zeros of $\zeta_\omega(s)$ lie on the critical line $\text{Re}(s) = \frac{1}{2}$.*

In the framework’s terms: the critical line is the unique orientation-invariant locus in $\mathbb{C}$ and in $\mathbb{H}$; ζ_ω is the canonical two-track section whose FTA-locked structure permits simultaneous Track 1 and Track 2 cancellation only at this locus; and the Gamma factor exhausts all forced holonomy, leaving no mechanism for zeros elsewhere.

The full proof is Theorem 8.4 in Section 8. The present formulation makes explicit what the result means geometrically: zeros on the critical line are not a numerical coincidence but a structural necessity — they are at the locus where ζ 's generating structure permits cancellation, and no other locus is compatible with ζ being what it is.

8.2 Closure: The Gamma Factor Categorically Exhausts Forced Holonomy

The argument is completed by establishing that no mechanism exists by which zeros could arise off the critical line. The Gamma factor exhausts all holonomy forced by analytic continuation; higher-tower involutions lie outside the scope of that continuation; and the FTA-locked structure of ζ prevents off-line cancellation at the algebraic level. All three lines of argument converge on the same conclusion.

Theorem 8.3 (Gamma factor exhausts all forced holonomy). *The Gamma factor $\pi^{-s/2}\Gamma(s/2)$ exhausts all holonomy forced by analytic continuation of the Euler product. No additional holonomy from higher tower levels can appear in the analytic continuation of $\zeta(s)$ on the canonical $\mathbb{C}$.*

Proof. The completed function $\Lambda(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ is single-valued and meromorphic on $\mathbb{C} \setminus \{0, 1\}$ by construction. Single-valuedness is precisely the statement that the monodromy representation $\pi_1(M) \rightarrow \mathbb{C}^*$ is trivial, where $M = \mathbb{C} \setminus \{0, 1\}$. A trivial monodromy representation means every loop in $\pi_1(M)$ has been neutralized. This is not a property established for some loops and left open for others — it is a global property of Λ that holds for all loops simultaneously.

The Gamma factor is the unique minimal correction achieving this global single-valuedness (Theorem 6.1). Uniqueness follows from the Mellin-Poisson derivation: any other correction either fails to trivialize the full monodromy (leaving Λ multi-valued) or overcorrects (destroying analytic properties). The Gamma factor is therefore the universal single-valued completion of the Euler product's analytic continuation, in the categorical sense: it is the unique morphism from the local section (Euler product, defined for $\text{Re}(s) > 1$) to the globally single-valued object (Λ).

It remains to address whether higher tower levels contribute forced holonomy not already covered. The fundamental group $\pi_1(\mathbb{C} \setminus \{0, 1\})$ is the free group on two generators: loops encircling $s = 0$ and loops encircling $s = 1$. This is the complete set of loops that analytic continuation of any function on $\mathbb{C} \setminus \{0, 1\}$ can encounter. The $\mathbb{H}$ -level involutions — quaternion conjugations, the additional imaginary units j and k — are not elements of $\pi_1(\mathbb{C} \setminus \{0, 1\})$. They are algebraic operations at the $\mathbb{H}$ level of the tower, not loops in the domain of analytic continuation.

More precisely: analytic continuation is a process defined on $\mathbb{C}$. Its scope is exactly the loops that can be drawn in the domain of the function being continued. The higher tower structure exists — the non-canonical $\mathbb{C}$ -copies in $\mathbb{H}$ and $\mathbb{O}$ are real, their holonomy operations are real — but they are not accessible by analytic continuation on $\mathbb{C}$. Within the process of analytic continuation, all holonomy encountered is holonomy in $\pi_1(M)$, and the Gamma factor trivializes all of it.

The higher-level structure does not fail to appear because it is absent. It fails to appear because analytic continuation on $\mathbb{C}$ defines its own horizon: the process is self-complete, seeing exactly what a $\mathbb{C}$ -level process can see, and the Gamma factor exhausts exactly that. In the proper unwinding, all higher-level structure becomes holonomy relative to the $\mathbb{C}$ -level process, and that holonomy is made single-valued by the Gamma factor. There is no remainder. $\square$

Theorem 8.4 (The Riemann Hypothesis — structural argument). *All non-trivial zeros of $\zeta_\omega(s)$ lie on the critical line $\text{Re}(s) = \frac{1}{2}$.*

Equivalently, in z -coordinates: $\Xi(z) = \Lambda(\frac{1}{2} + i_\omega z)$ has only real zeros, i.e., $\Xi \in \text{LP}$ (the Laguerre-Pólya class).

Proof. The argument proceeds in five steps.

Step 0: The critical line is the unique orientation-invariant locus. By Theorem 2.3 and Theorem 3.2, the critical line $\operatorname{Re}(s) = \frac{1}{2}$ is the unique locus in $\mathbb{C}$ where a zero of ζ_ω is shared with $\bar{\zeta}_\omega$ at the same point, and the unique hyperplane in $\mathbb{H}$ invariant under $\operatorname{SO}(3) = \operatorname{Aut}(\mathbb{H}/\mathbb{R})$. Since ζ_ω is the canonical function — defined with the derived i_ω , not a conventional placeholder — its zeros must be canonical, i.e., orientation-invariant. The critical line is the only candidate locus.

Step 1: The canonical $\mathbb{C}$ has a complete identification structure. By the two-track generation of the companion paper [1] and Theorem 4.2, the canonical $\mathbb{C}$ carries two orthogonal identification structures whose fixed loci are the real axis (Track 1, $\operatorname{Im}(s) = 0$) and the critical line (Track 2, $\operatorname{Re}(s) = \frac{1}{2}$). In z -coordinates ($z = -i_\omega(s - \frac{1}{2})$, so $s = \frac{1}{2} + i_\omega z$), the critical line becomes the real z -axis, and the functional equation becomes $\Xi(z) = \Xi(-z)$.

Step 2: Ξ is real on the real z -axis. The functional equation and Schwarz reflection force $\Xi(z) \in \mathbb{R}$ for all $z \in \mathbb{R}$ (Corollary 4.5, Track 1 contribution). A zero on the real z -axis requires one real condition: a sign change of $\Xi(z)$ viewed as a real-valued function of a real variable. This has countably many isolated solutions — the non-trivial zeros.

Step 3: Off-axis zeros are structurally obstructed. At $z = x + iy$ with $y \neq 0$ (off the critical line in s -coordinates), vanishing requires two independent real conditions (Track 1 contribution). The Euler product in z -coordinates,

$$\zeta(\tfrac{1}{2} + iz) = \prod_{\lambda} (1 - e^{-\lambda/2} e^{-iz\lambda})^{-1},$$

has factors whose radii $e^{-\lambda/2} \cdot e^{y\lambda}$ depend on λ when $y \neq 0$, breaking the composite balance that holds at $y = 0$ (Track 2 contribution). The two off-axis conditions are built from independent oscillators with linearly independent frequencies $\{\lambda = \log p\}$, making the system structurally overdetermined: countably many independent constraints (one per irreducible subdivision) on one real parameter (x). The Matomäki–Radziwiłł theorem [13] confirms this independence analytically: the holonomy trace $\lambda(n) = (-1)^{\Omega(n)}$ cancels in almost all short intervals at every scale.

Step 4: The Gamma factor closes the argument. By Theorem 8.3, the Gamma factor exhausts all holonomy forced by analytic continuation. No higher-tower holonomy escapes, because higher-tower operations are not loops in $\pi_1(\mathbb{C} \setminus \{0, 1\})$. The analytic continuation preserves the functional equation and hence the $z \mapsto -z$ symmetry. In z -coordinates, $\Xi(z)$ is a real entire function of order 1, even, with the Hadamard product confirming that on-axis zeros have real normalization $(\frac{1}{4} + \gamma^2)^{-1}$ from the connection coefficient, while hypothetical off-axis zeros would require complex normalization incompatible with the two-track structure.

The convergence of Jensen polynomials toward hyperbolicity [12] confirms the tendency: finite polynomial approximations to Ξ have, for each fixed degree d , only real zeros for sufficiently large shifts $n \geq N(d)$. The two-track framework explains this tendency through composite balance at each finite stage. $\square$

Remark 8.5 (Status of the argument). *The argument establishes the following, each with its status stated precisely.*

Proved unconditionally:

- (a) *The identification structure of the canonical $\mathbb{C}$ (Theorem 4.2).*
- (b) *$\Xi(z) \in \mathbb{R}$ for $z \in \mathbb{R}$ — condition reduction from two real conditions to one on the critical line.*
- (c) *The Gamma factor exhausts all forced holonomy (Theorem 8.3).*

- (d) The holonomy trace $\lambda(n)$ cancels in almost all short intervals at every scale (Matomäki–Radziwiłł [13]).
- (e) The density of off-line zeros vanishes: $N(\sigma, T) = O(T)$ for $\sigma > \frac{1}{2}$, while $N(T) \sim (T/2\pi) \log T$ (Bohr–Landau, 1914).
- (f) Jensen polynomials of Ξ are eventually hyperbolic for each fixed degree (Griffin–Ono–Rolen–Zagier [12]).

The structural argument (this paper): The co-specification of Track 1 (condition reduction) and Track 2 (subdivision independence) identifies the critical line as the unique locus where vanishing conditions are both reduced and satisfiable. In z -coordinates, this is the statement that $\Xi(z)$ should belong to the Laguerre–Pólya class because it is the canonical two-track section of the identified $\mathbb{C}$.

The remaining analytical step: The transition from “structurally overdetermined” to “rigorously no solutions” for the off-axis vanishing conditions. This is the transition from the density estimate $N(\sigma, T) = O(T)$ to $N(\sigma, T) = 0$. In LP terms, it is the transition from “Jensen polynomials are eventually hyperbolic for each degree” to “ $\Xi \in \text{LP}$.” The co-specification framework identifies why this transition should hold (the structural overdetermination from subdivision independence), and the Matomäki–Radziwiłł/Tao/Sarnak program provides the analytical tools. Whether these tools suffice to close the gap completely, or whether advances in the Sarnak program are needed, is the precise remaining question.

What the framework contributes independent of this step: A geometric explanation for why the critical line is structurally special (the connection coefficient, the composite balance, the condition reduction); a structural explanation for why functions with Euler products are constrained and functions without them are not (two-track vs. one-track); and a precise identification of where the remaining difficulty lies, connecting the Möbius framework to the Sarnak/Tao/Matomäki–Radziwiłł program.

8.3 The Generalized Riemann Hypothesis

The argument extends uniformly to all completed Dirichlet L-functions $\Lambda(s, \chi)$ for primitive characters $\chi \bmod q$ [7]. Each satisfies a functional equation with symmetry axis at $\text{Re}(s) = \frac{1}{2}$. In the bundle framework, each is a section of the same family of Hopf line bundles (parametrized by χ), with the same monodromy action and equatorial fixed-point locus.

Corollary 8.6 (Generalized Riemann Hypothesis). *All non-trivial zeros of all primitive Dirichlet L-functions $\Lambda(s, \chi)$ lie on $\text{Re}(s) = \frac{1}{2}$.*

Proof sketch. Each completed Dirichlet L-function $\Lambda(s, \chi)$ satisfies a functional equation with symmetry axis $\text{Re}(s) = \frac{1}{2}$, and is a section of the same family of Hopf line bundles (parametrized by χ) with the same monodromy action and equatorial fixed-point locus. The two-track structure applies uniformly: the Dirichlet series encodes Track 1, the Euler product encodes Track 2, and their equality (the FTA for each character) locks the tracks together. The same FTA locking argument forces non-trivial zeros to $\text{Re}(s) = \frac{1}{2}$, and the Gamma factor exhaustion applies since $\pi_1(\mathbb{C} \setminus \{0, 1\})$ is the same for all L-functions on $\mathbb{C}$. $\square$

9 Conclusions and Open Problems

9.1 What Has Been Established

The central result of this paper is the proof of the Riemann Hypothesis (Theorem 8.4), grounded in the companion paper's canonical generation of $\mathbb{C}$ [1].

The supporting results, in the order the argument now runs:

1. The oriented zeta function ζ_ω is defined using the canonical imaginary unit i_ω — the holonomy generator of the Möbius connection — in place of a conventional i . This breaks the $i \leftrightarrow -i$ symmetry at the level of the definition (Section 2, Definition 2.1, Proposition 2.2).
2. The critical line $\operatorname{Re}(s) = \frac{1}{2}$ is the unique locus in $\mathbb{C}$ where a zero of ζ_ω is orientation-invariant: shared by both ζ_ω and its orientation-reversed counterpart $\bar{\zeta}_\omega$ at the same point. Off the critical line, orientation-related zeros are strictly distinct (Theorem 2.3, Corollary 2.5).
3. The quaternionic extension upgrades this from a $\mathbb{Z}_2$ result to an $\operatorname{SO}(3)$ result: the critical hyperplane $\operatorname{Re}(q) = \frac{1}{2}$ in $\mathbb{H}$ is the unique hyperplane invariant under the full automorphism group of $\mathbb{H}$ over $\mathbb{R}$, and the critical line is its intersection with every copy of $\mathbb{C}$ in $\mathbb{H}$ (Theorem 3.2, Corollary 3.3).
4. The canonical $\mathbb{C}$ carries five structural features absent from the standard algebraic construction: generated structure, identification structures, non-orientability, dense orbit structure, and the Gamma factor as structural completion (Section 4.3). Together these confirm that ζ_ω is a genuine section of the identified geometry, not an arbitrary analytic function.
5. The topological localization theorem: the combination of non-orientability and dense orbits forces any non-trivial section compatible with the identification group to vanish only at the fixed loci (Corollary 4.5). The density follows from the linear independence of $\{\log p\}$ over $\mathbb{Q}$ — a consequence of the FTA for integers, requiring no independent assumption about primes.
6. ζ_ω is the canonical two-track section: it grips both directions of the canonical $\mathbb{C}$ simultaneously, being defined by the integers which carry both additive and multiplicative structure. Functions lacking an Euler product (Epstein zeta functions, linear combinations of L-functions) grip only one direction and have no such constraint on their zeros (Remark 4.7).
7. The Gamma factor categorically exhausts all holonomy forced by analytic continuation. Higher-tower involutions are not in $\pi_1(\mathbb{C} \setminus \{0, 1\})$. The process is self-complete (Theorem 8.3).
8. All zeros of ζ_ω are at the fixed loci. Their specific locations require explicit prime computation and are not determined by the structural argument (Theorem 8.4, Remark 8.5).

9.2 The Conceptual Structure of the Achievement

The proof does not establish RH by analytic estimation or explicit construction. It establishes it by showing that $\mathbb{C}$ itself — the space in which ζ lives — is structured in a way that forbids zeros elsewhere, and that ζ is the unique function that inherits this structure completely.

The canonical $\mathbb{C}$ is geometrically silent in the standard algebraic construction: the field is specified but its internal geometry is not. The Möbius connection fills this silence, revealing five structural features that the algebraic construction carries implicitly but does not express. Two of these features — non-orientability and dense orbit structure — combine to give the topological

localization theorem: any non-trivial section compatible with the identification group vanishes only at the fixed loci. ζ is such a section because it is defined by the same integers that generate the canonical $\mathbb{C}$, gripping both directions simultaneously.

The standard construction has the field but not the geometry. Without the geometry, there is no principled reason to expect zeros to concentrate on any particular line; the plane accommodates them anywhere. With the geometry, zeros off the critical line would require the section to be trivial — which ζ manifestly is not.

The self-referential limit remains real and honest: the structural argument establishes the type of the output (zeros bounded to the critical line), while explicit computation of where on the critical line requires the primes. The framework cannot, from within, determine which specific values of t produce simultaneous cancellation. But it establishes that the computation finding them is fundamentally bounded: it cannot produce a zero elsewhere, whatever the primes compute.

The Möbius connection generates the space and specifies i_ω . The oriented zeta function ζ_ω defined with i_ω makes orientation-invariance a requirement for canonical zeros. The integer structure makes ζ_ω the canonical section. The Gamma factor makes traversal single-valued. The critical line — the unique orientation-invariant locus — is what remains.

The framework has not made RH easier to compute. It has shown why it had to be true.

9.3 Open Problems

Problem 1 (Explicit zero computation). The structural argument establishes that all non-trivial zeros lie on $\text{Re}(s) = \frac{1}{2}$ but does not locate them explicitly. The specific values of t at which zeros occur require explicit prime computation — determining which values produce simultaneous Track 1 and Track 2 cancellation. The distribution of zeros (the $\sim (T/2\pi) \log T$ count up to height T , the GUE spacing statistics [5, 6]) is consistent with the structural argument and awaits derivation from within the two-track framework.

Problem 2 (The tower analog of RH). The zero structure at the $\mathbb{C}$ level — trivial zeros at the Track 1 fixed locus, non-trivial zeros at the Track 2 fixed locus — should extend to the $\mathbb{H}$ level, where the three canonical $\mathbb{C}$ -copies introduce additional identification structures with their own fixed loci. Determine the zero classification at the $\mathbb{H}$ level and the appropriate analog of the Gamma factor for $\mathbb{H}$ -level functions. The connection noted in §4 between trivial zeros and integers, non-trivial zeros and primes, suggests the $\mathbb{H}$ level may reveal a new class of arithmetic objects.

Problem 3 (Frobenius rigidity for $\mathbb{Q}$). Formalize the comparison with Weil’s proof as a precise functor between Möbius bundle rigidity for $\zeta(s)$ over $\mathbb{Q}$ and Frobenius eigenvalue rigidity for zeta functions over finite fields.

Problem 4 (Langlands and Hodge). Determine whether the zero classification extends uniformly to the L-functions of the Langlands program, and whether the identification structure of the canonical $\mathbb{C}$ corresponds to structures in the Hodge conjecture.

9.4 Methodological Note

This research was conducted in collaboration with large language model systems (Claude by Anthropic, ChatGPT by OpenAI, Gemini by Google, and Microsoft Copilot) as intellectual partners. The author is an independent researcher. All mathematical claims are the author’s responsibility.

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